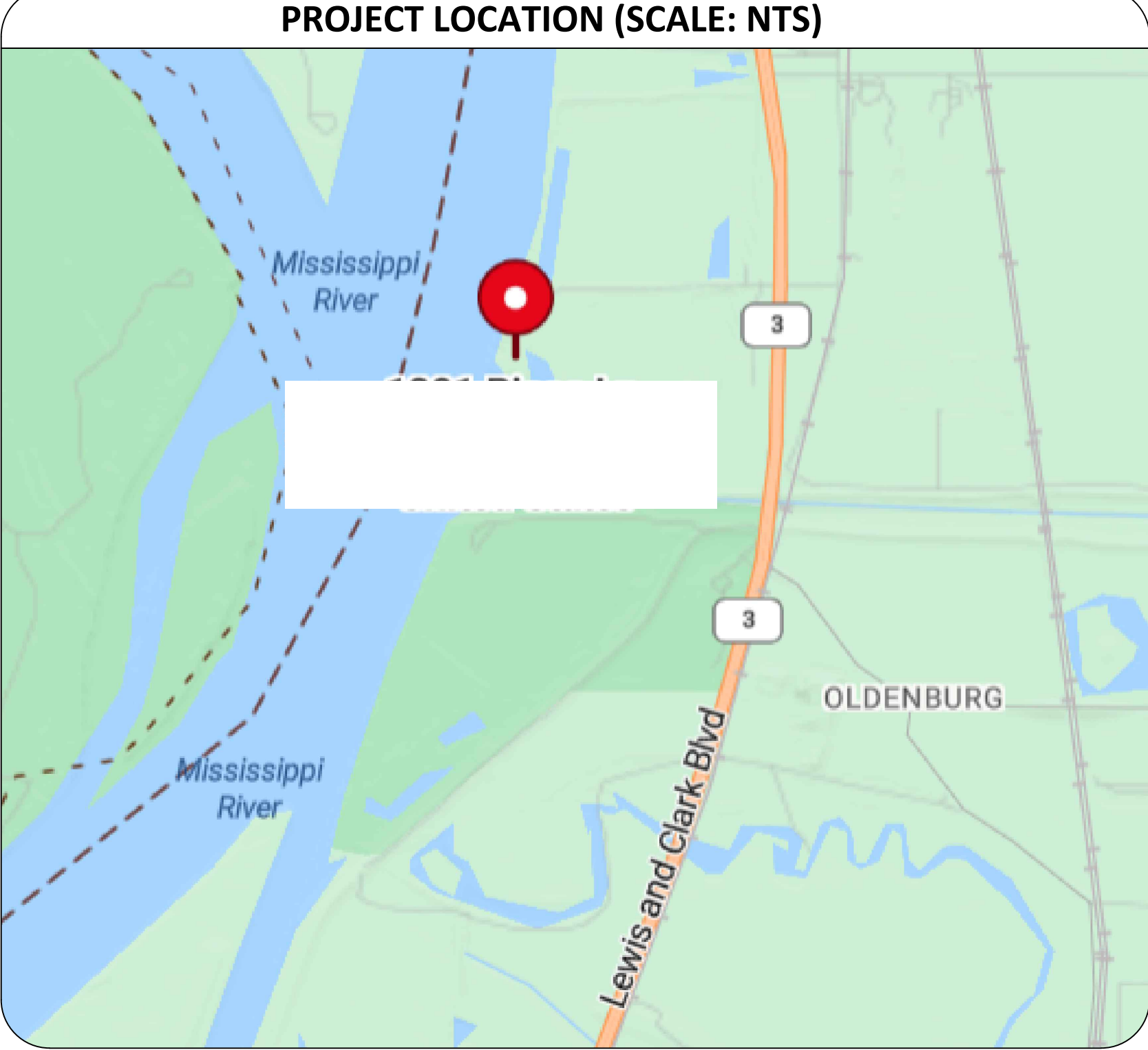
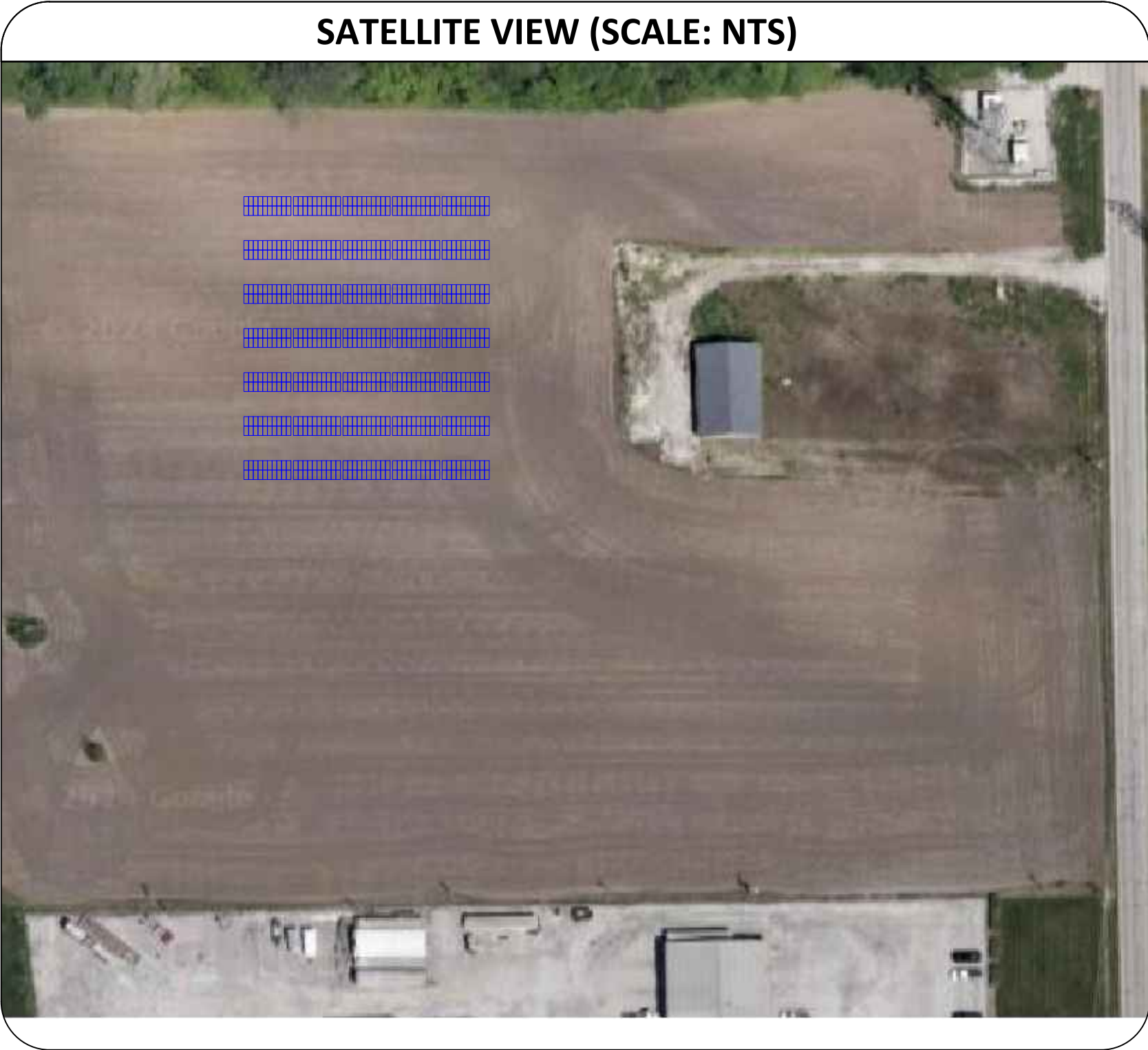


| PROJECT: PV SYSTEM DETAILS |                         |
|----------------------------|-------------------------|
| [ DC ] SYSTEM SIZE         | [ AC ] SYSTEM SIZE      |
| (700) MODULES x 575W       | (14) INVERTERS x 25000W |
| 402.50 kW                  | 350.00 kW               |



| SCOPE OF WORK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>PROPOSED PROJECT TO INSTALL GRID- TIED PV ONLY SYSTEM ON AN EXISTING GROUND MOUNT STRUCTURE AT A COMMERCIAL LOCATION: 1234 STREET NAME, CITY, IL 9874. COORDINATES: 38.811044, -90.111083</li><li>THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM.</li><li>INSTALLED PV SYSTEM WILL BE THEN CONNECTED TO HOMEOWNER'S SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ).</li><li>THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE HOME APPLIANCES. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.</li></ul> |

| APPLICABLE GOVERNING CODES & NOTES                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES:                                                                                                                                                     |
| <ul style="list-style-type: none"><li>2021 INTERNATIONAL BUILDING CODE</li><li>2021 INTERNATIONAL FIRE CODE</li><li>2021 INTERNATIONAL RESIDENTIAL CODE</li><li>2020 NATIONAL ELECTRICAL CODE</li></ul> |
| ALL OTHER RELATED STATE & AHJ BY-LAWS & ORDINANCES                                                                                                                                                      |

| DESIGN CRITERIA                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>CONSTRUCTION: COMMERCIAL</li><li>EXPOSURE CATEGORY : C, RISK CATEGORY : I</li><li>WIND SPEED (ASCE 7-10) : 115MPH</li><li>GROUND SNOW LOAD (ASCE 7-10) : 20 PSF</li><li>ASHRAE EXTREME LOW: -19°C</li><li>ASHRAE HIGH TEMP (2% AVG.) : 35°C</li><li>ASHRAE DISTANCE ABOVE ROOF (7/8") : 57°C</li></ul> |

| SHEET INDEX |                             |
|-------------|-----------------------------|
| PV-0        | PROJECT SUMMARY             |
| PV-1        | SITE PLAN                   |
| PV-2        | ROOF ATTACHMENT DETAILS     |
| PV-3        | BOM & STRING LAYOUT         |
| S-1         | STRUCTURAL DETAILED DIAGRAM |
| E-1 & E-4   | ELECTRICAL DIAGRAM          |
| E-5 & E-6   | ELECTRICAL CALCULATIONS     |
| E-7 & E-8   | PLACARDS                    |
| E.S         | EQUIPMENT SPECIFICATIONS    |

| EQUIPMENT DETAILS   |                                                                     |
|---------------------|---------------------------------------------------------------------|
| (#) PV MODULES      | (700) VSUN570N-144BMH-DG                                            |
| (#)INVERTER         | (14) SOLECTRIA PVI-25TL-208 INVERTER (208 V)                        |
| OUTPUT CURRENT      | 973.0A                                                              |
| ROOF RACKING SYSTEM | UNIRAC GFT W/ GFT RAIL 246" MILL                                    |
| BRANCH CIRCUIT      | 28 STRINGS OF 13 MODULES & 28 STRINGS OF 12 MODULES                 |
| BRANCH CIRCUIT      | LINE SIDE TAP IN THE CT CABINET , 1200A BUSBAR AND 120/240V, 3W, 1ϕ |
| UTILITY             | -                                                                   |
| AHJ                 | CHOUTEAU TOWNSHIP                                                   |

| GENERAL NOTES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>ALL COMPONENTS ARE UL LISTED &amp; NEC CERTIFIED, WHERE WARRANTED</li><li>A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS.</li><li>HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER NEC CODE 240.24.</li><li>A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH NEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER NEC GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM.</li><li>PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE.</li><li>PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS.</li><li>WATER PROOF CONNECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER NEC 312.2 AND 314.15.</li><li>GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER NEC 250.92.</li><li>ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE.</li><li>CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS &amp; NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM.</li><li>CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS &amp; DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.</li></ul> |
| PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| ELECTRICAL NOTES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONNECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (NEC690.4(E) AND 705.6)</li><li>LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONNECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF.</li><li>NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES.</li><li>ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL.</li><li>ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS .</li><li>WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING</li><li>BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES.</li><li>ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT.</li><li>CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO NEC.</li><li>MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT. THE VOLTAGES FOR THE 240VAC RATED.</li></ul> |

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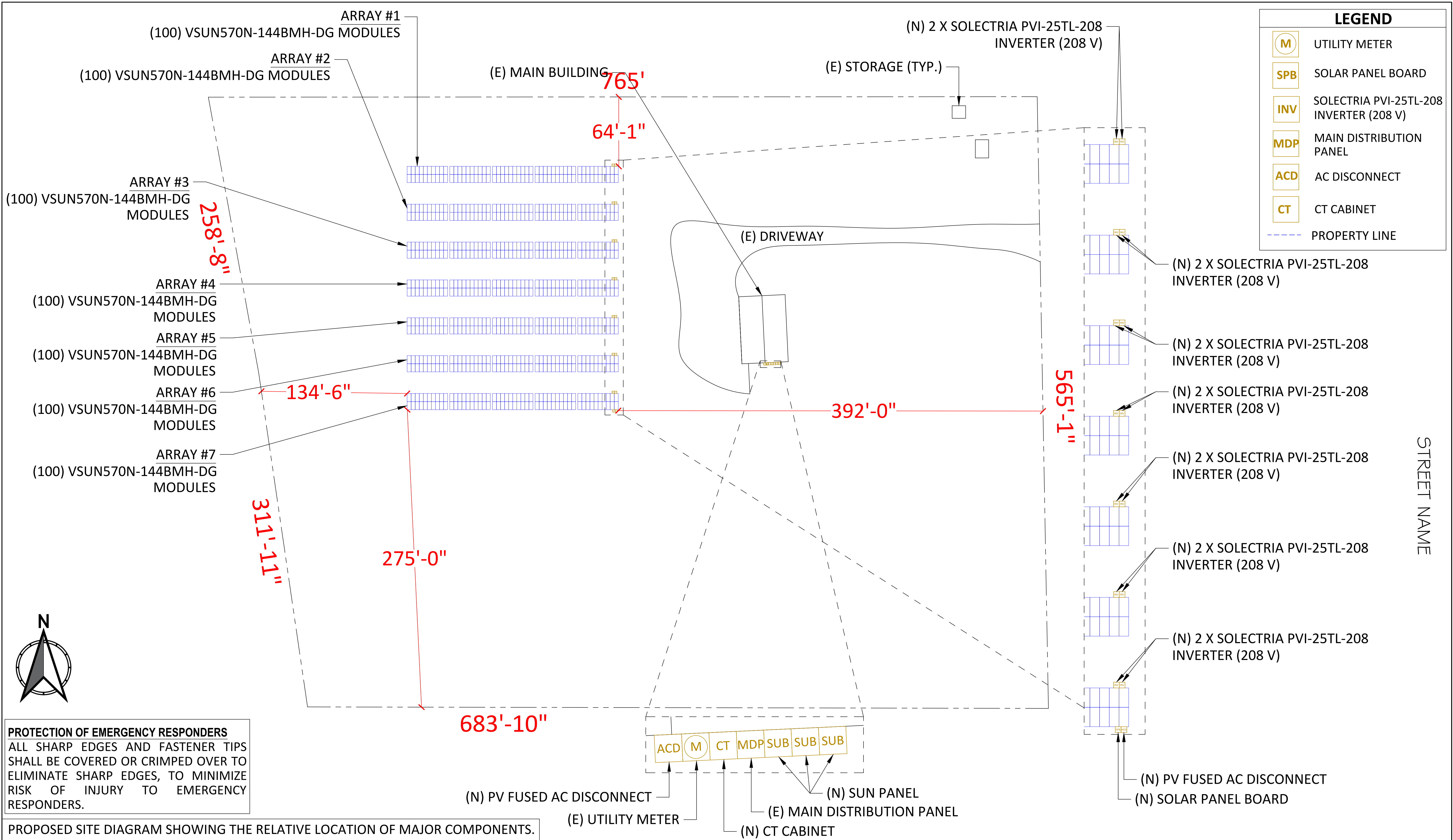
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|------------|----------|
| DATE       |          |
| PROJECT ID |          |
| CREATED BY | AK       |
| CHECKED BY | MSB      |
| SCALE      | NTS      |
| SHEET SIZE | 24"X 36" |
|            | ANSI D   |

|                                 |  |
|---------------------------------|--|
| GROUND MOUNT COMMERICAL         |  |
| 1234 STREET NAME, CITY, IL 9874 |  |
| APN: ----                       |  |
| HOMEOWNER INFO                  |  |

| # | REVISION/UPDATE | DATE |
|---|-----------------|------|
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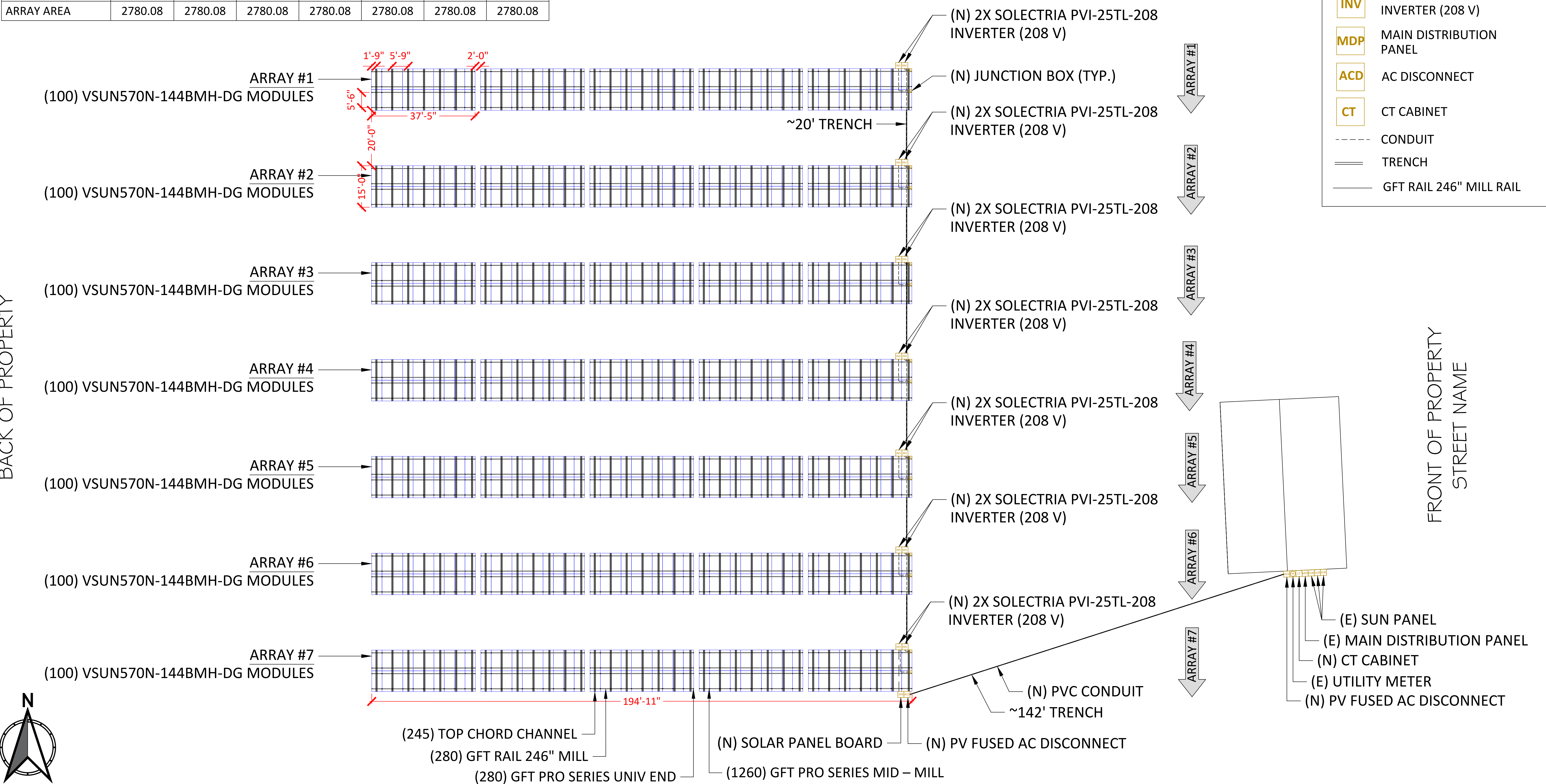
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|                   |
| SIGNED & STAMP BY |

|                   |
|-------------------|
| PV-0              |
| SHEET NAME        |
| PROJECT SUMMARY   |
| SHEET INFORMATION |



|            |            |                |                                                                                |            |                 |                   |           |                           |               |                |               |                |          |                   |           |                   |           |                   |           |        |        |  |  |                   |  |                   |                   |
|------------|------------|----------------|--------------------------------------------------------------------------------|------------|-----------------|-------------------|-----------|---------------------------|---------------|----------------|---------------|----------------|----------|-------------------|-----------|-------------------|-----------|-------------------|-----------|--------|--------|--|--|-------------------|--|-------------------|-------------------|
|            | DATE       |                | <b>GROUND MOUNT COMMERICAL</b><br>1234 STREET NAME, CITY, IL 9874<br>APN: ---- | #          | REVISION/UPDATE | DATE              |           | <b>PV-1</b><br>SHEET NAME |               |                |               |                |          |                   |           |                   |           |                   |           |        |        |  |  |                   |  |                   |                   |
|            | PROJECT ID |                |                                                                                | CREATED BY | AK              | CHECKED BY        |           |                           | MSB           | SCALE          | 1/38" = 1'-0" | SHEET SIZE     | 24"X 36" | HOMEOWNER INFO    |           |                   |           | SIGNED & STAMP BY | SITE PLAN |        | ANSI D |  |  |                   |  |                   | SHEET INFORMATION |
|            | CREATED BY | AK             |                                                                                | CHECKED BY | MSB             | SCALE             |           |                           | 1/38" = 1'-0" | SHEET SIZE     | 24"X 36"      | HOMEOWNER INFO |          |                   |           | SIGNED & STAMP BY | SITE PLAN |                   |           | ANSI D |        |  |  |                   |  | SHEET INFORMATION |                   |
|            | CHECKED BY | MSB            |                                                                                | SCALE      | 1/38" = 1'-0"   | SHEET SIZE        |           |                           | 24"X 36"      | HOMEOWNER INFO |               |                |          | SIGNED & STAMP BY | SITE PLAN |                   |           | ANSI D            |           |        |        |  |  | SHEET INFORMATION |  |                   |                   |
|            | SCALE      | 1/38" = 1'-0"  |                                                                                |            |                 |                   |           |                           |               |                |               |                |          |                   |           |                   |           |                   |           |        |        |  |  |                   |  |                   |                   |
| SHEET SIZE | 24"X 36"   | HOMEOWNER INFO |                                                                                |            |                 | SIGNED & STAMP BY | SITE PLAN |                           |               |                |               |                |          |                   |           |                   |           |                   |           |        |        |  |  |                   |  |                   |                   |
|            | ANSI D     |                |                                                                                |            |                 |                   |           | SHEET INFORMATION         |               |                |               |                |          |                   |           |                   |           |                   |           |        |        |  |  |                   |  |                   |                   |

| GROUND MOUNT ARRAY DETAILS |          |          |          |          |          |          |          |
|----------------------------|----------|----------|----------|----------|----------|----------|----------|
| MOUNTING PLANE             | ARRAY #1 | ARRAY #2 | ARRAY #3 | ARRAY #4 | ARRAY #5 | ARRAY #6 | ARRAY #7 |
| NO. OF MODULES             | 100      | 100      | 100      | 100      | 100      | 100      | 100      |
| AZIMUTH                    | 180°     | 180°     | 180°     | 180°     | 180°     | 180°     | 180°     |
| ARRAY TILT                 | 30°      | 30°      | 30°      | 30°      | 30°      | 30°      | 30°      |
| ARRAY AREA                 | 2780.08  | 2780.08  | 2780.08  | 2780.08  | 2780.08  | 2780.08  | 2780.08  |



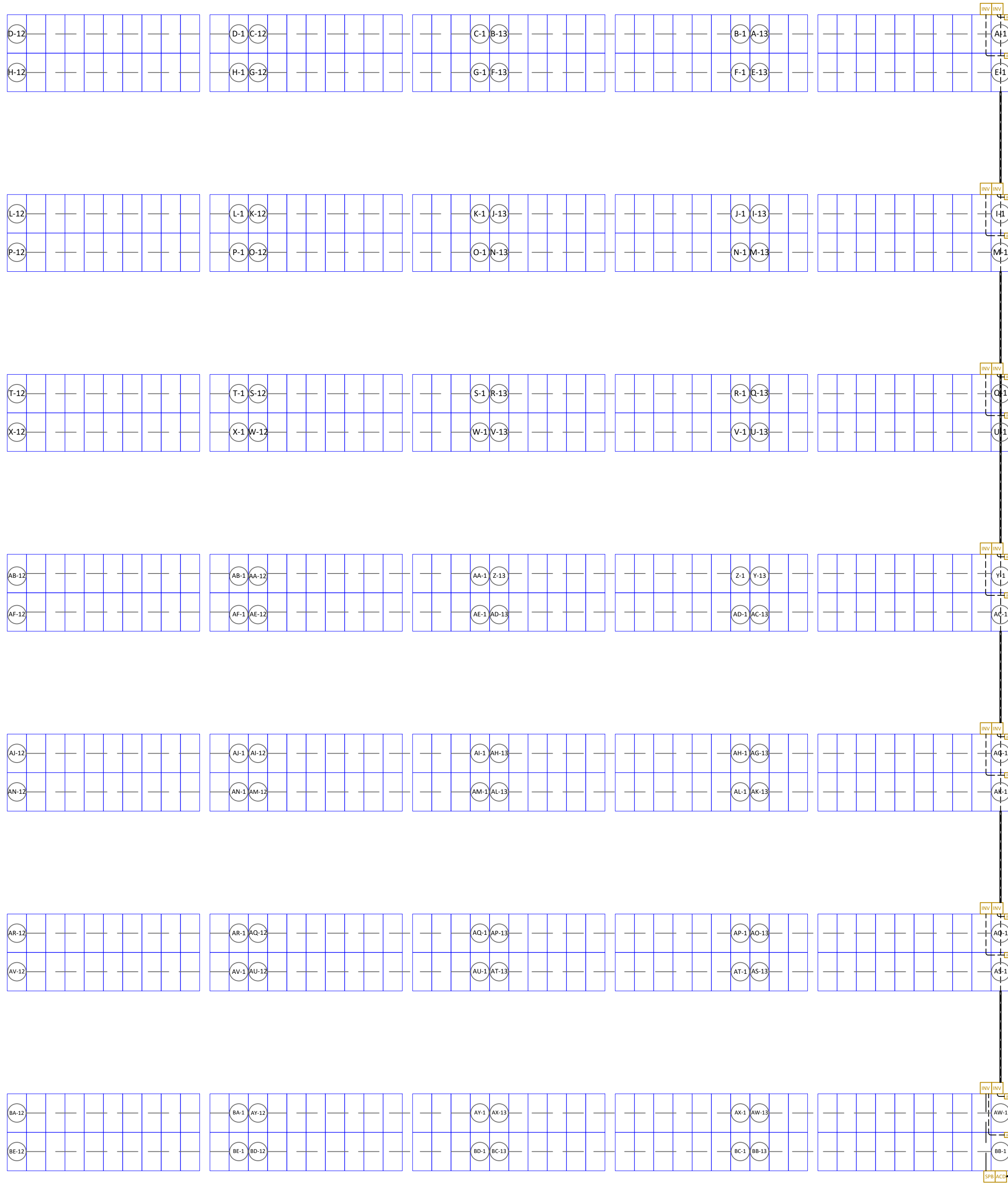
|            |            |               |                                                                                |            |                 |            |  |                              |                                                                      |               |            |                |            |                |        |                |
|------------|------------|---------------|--------------------------------------------------------------------------------|------------|-----------------|------------|--|------------------------------|----------------------------------------------------------------------|---------------|------------|----------------|------------|----------------|--------|----------------|
|            | DATE       |               | <b>GROUND MOUNT COMMERCIAL</b><br>1234 STREET NAME, CITY, IL 9874<br>APN: ---- | #          | REVISION/UPDATE | DATE       |  | <b>SIGNED &amp; STAMP BY</b> | <b>PV-2</b><br>SHEET NAME<br>ATTACHMENT DETAILS<br>SHEET INFORMATION |               |            |                |            |                |        |                |
|            | PROJECT ID |               |                                                                                | CREATED BY | AK              | CHECKED BY |  |                              |                                                                      | MSB           | SCALE      | 1/16" = 1'-0"  | SHEET SIZE | 24"X 36"       | ANSI D | HOMEOWNER INFO |
|            | CREATED BY | AK            |                                                                                | CHECKED BY | MSB             | SCALE      |  |                              |                                                                      | 1/16" = 1'-0" | SHEET SIZE | 24"X 36"       | ANSI D     | HOMEOWNER INFO |        |                |
|            | CHECKED BY | MSB           |                                                                                | SCALE      | 1/16" = 1'-0"   | SHEET SIZE |  |                              |                                                                      | 24"X 36"      | ANSI D     | HOMEOWNER INFO |            |                |        |                |
|            | SCALE      | 1/16" = 1'-0" |                                                                                |            |                 |            |  |                              |                                                                      |               |            |                |            |                |        |                |
| SHEET SIZE | 24"X 36"   | ANSI D        | HOMEOWNER INFO                                                                 |            |                 |            |  |                              |                                                                      |               |            |                |            |                |        |                |

| BILL OF MATERIALS |      |                                                                      |
|-------------------|------|----------------------------------------------------------------------|
| EQUIPMENT         | QTY  | DESCRIPTION                                                          |
| SOLAR PV MODULE   | 700  | VSUN570N-144BMH-DG                                                   |
| INVERTER          | 14   | SOLECTRIA PVI-25TL-208 INVERTER (208 V)                              |
| JUNCTION BOX      | 14   | JUNCTION BOX, 600V, NEMA 3R, UL LISTED                               |
| AC DISCONNECT     | 2    | 1200A PV FUSED AC DISCONNECT, (2) 1200A FUSES, 120/240V, 2P, NEMA 3R |
| SOLAR BOARD PANEL | 1    | 1000A SOLAR PANEL BOARD, NEMA 3R, 120/240V, UL LISTED                |
| SUB PANEL         | 3    | 400A SUB PANEL, NEMA 3R, 120/240V, UL LISTED                         |
| RAIL              | 280  | GFT RAIL 246" MILL                                                   |
| BONDED SPLICE     | 140  | GFT RAIL SPLICE KIT                                                  |
| FOUNDATION        | 245  | GFT C-PILE , 15'                                                     |
| STRUCTURE         | 245  | GFT TOP CHORD CHANNEL 20/30 LGFMT                                    |
| STRUCTURE         | 245  | GFT DIAGONAL BRACE ASSEMBLY 30D SR                                   |
| HARDWARE          | 245  | KIT, GFT ASSEMBLY HARDWARE 4-RAIL                                    |
| WIRE MANNAGEMENT  | 385  | GFT WIRE MANAGEMENT CLIP                                             |
| WIRE MGMT         | 2800 | HEYCO SUNRUNNER 2-S CLIP                                             |
| MID CLAMP         | 1260 | GFT PRO SERIES MID – MILL                                            |
| END CLAMP         | 280  | GFT PRO SERIES UNIV END                                              |
| GROUNDING LUG     | 35   | ILSCO LAY IN LUG (GBL4DBT)                                           |

| STRING LEGEND |                         |    |                         |
|---------------|-------------------------|----|-------------------------|
| A             | STRING #1 - 13 MODULES  | R  | STRING #15 - 13 MODULES |
| B             | STRING #2 - 13 MODULES  | S  | STRING #16 - 12 MODULES |
| C             | STRING #3 - 12 MODULES  | T  | STRING #17 - 12 MODULES |
| D             | STRING #4 - 12 MODULES  | U  | STRING #18 - 13 MODULES |
| E             | STRING #5 - 13 MODULES  | V  | STRING #19 - 13 MODULES |
| F             | STRING #6 - 13 MODULES  | W  | STRING #20 - 12 MODULES |
| G             | STRING #7 - 12 MODULES  | X  | STRING #21 - 12 MODULES |
| H             | STRING #8 - 12 MODULES  | Y  | STRING #22 - 13 MODULES |
| I             | STRING #9 - 13 MODULES  | Z  | STRING #23 - 13 MODULES |
| J             | STRING #10 - 13 MODULES | AA | STRING #24 - 12 MODULES |
| K             | STRING #11 - 12 MODULES | AB | STRING #25 - 12 MODULES |
| L             | STRING #12 - 12 MODULES | AC | STRING #26 - 13 MODULES |
| M             | STRING #13 - 13 MODULES | AD | STRING #26 - 13 MODULES |
| N             | STRING #13 - 13 MODULES | AE | STRING #26 - 12 MODULES |
| O             | STRING #13 - 12 MODULES | AF | STRING #26 - 13 MODULES |
| P             | STRING #13 - 12 MODULES | AG | STRING #1 - 13 MODULES  |
| Q             | STRING #14 - 13 MODULES | AH | STRING #2 - 13 MODULES  |

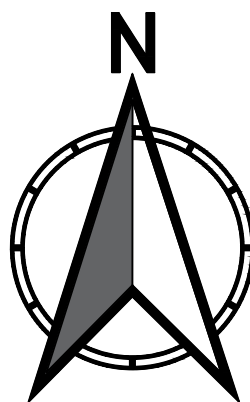
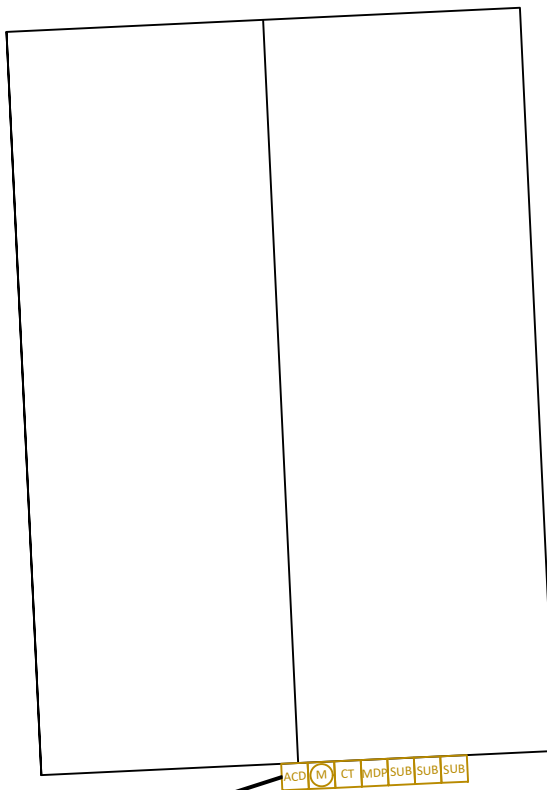
BACK OF PROPERTY

| STRING LEGEND |                         |               |                         |
|---------------|-------------------------|---------------|-------------------------|
| <div>AI</div> | STRING #2 - 12 MODULES  | <div>AO</div> | STRING #15 - 13 MODULES |
| <div>AJ</div> | STRING #2 - 12 MODULES  | <div>AP</div> | STRING #15 - 13 MODULES |
| <div>AK</div> | STRING #2 - 13 MODULES  | <div>AQ</div> | STRING #15 - 12 MODULES |
| <div>AL</div> | STRING #2 - 13 MODULES  | <div>AR</div> | STRING #15 - 12 MODULES |
| <div>AM</div> | STRING #14 - 12 MODULES | <div>AS</div> | STRING #1 - 13 MODULES  |
| <div>AN</div> | STRING #15 - 12 MODULES | <div>AT</div> | STRING #2 - 13 MODULES  |
| <div>AU</div> | STRING #2 - 12 MODULES  | <div>AV</div> | STRING #2 - 12 MODULES  |
| <div>AW</div> | STRING #2 - 13 MODULES  | <div>AX</div> | STRING #2 - 13 MODULES  |
| <div>AY</div> | STRING #14 - 12 MODULES | <div>AZ</div> | STRING #15 - 12 MODULES |
| <div>BA</div> | STRING #15 - 13 MODULES |               |                         |
| <div>BB</div> | STRING #15 - 13 MODULES |               |                         |
| <div>BC</div> | STRING #15 - 12 MODULES |               |                         |
| <div>BD</div> | STRING #15 - 12 MODULES |               |                         |
| <div>BE</div> | STRING #15 - 12 MODULES |               |                         |



| LEGEND           |                                         |
|------------------|-----------------------------------------|
| <div>M</div>     | UTILITY METER                           |
| <div>SPB</div>   | SOLAR PANEL BOARD                       |
| <div>INV</div>   | SOLECTRIA PVI-25TL-208 INVERTER (208 V) |
| <div>MDP</div>   | MAIN DISTRIBUTION PANEL                 |
| <div>ACD</div>   | AC DISCONNECT                           |
| <div>CT</div>    | CT CABINET                              |
| <div>=====</div> | TRENCH                                  |
| <div>-----</div> | CONDUIT                                 |

FRONT OF PROPERTY  
STREET NAME



|            |               |
|------------|---------------|
| DATE       |               |
| PROJECT ID |               |
| CREATED BY | AK            |
| CHECKED BY | MSB           |
| SCALE      | 1/10" = 1'-0" |
| SHEET SIZE | 24"X 36"      |
|            | ANSI D        |

GROUND MOUNT COMMERICAL

1234 STREET NAME, CITY, IL 9874

APN: ----

HOMEOWNER INFO

| # | REVISION/UPDATE | DATE |
|---|-----------------|------|
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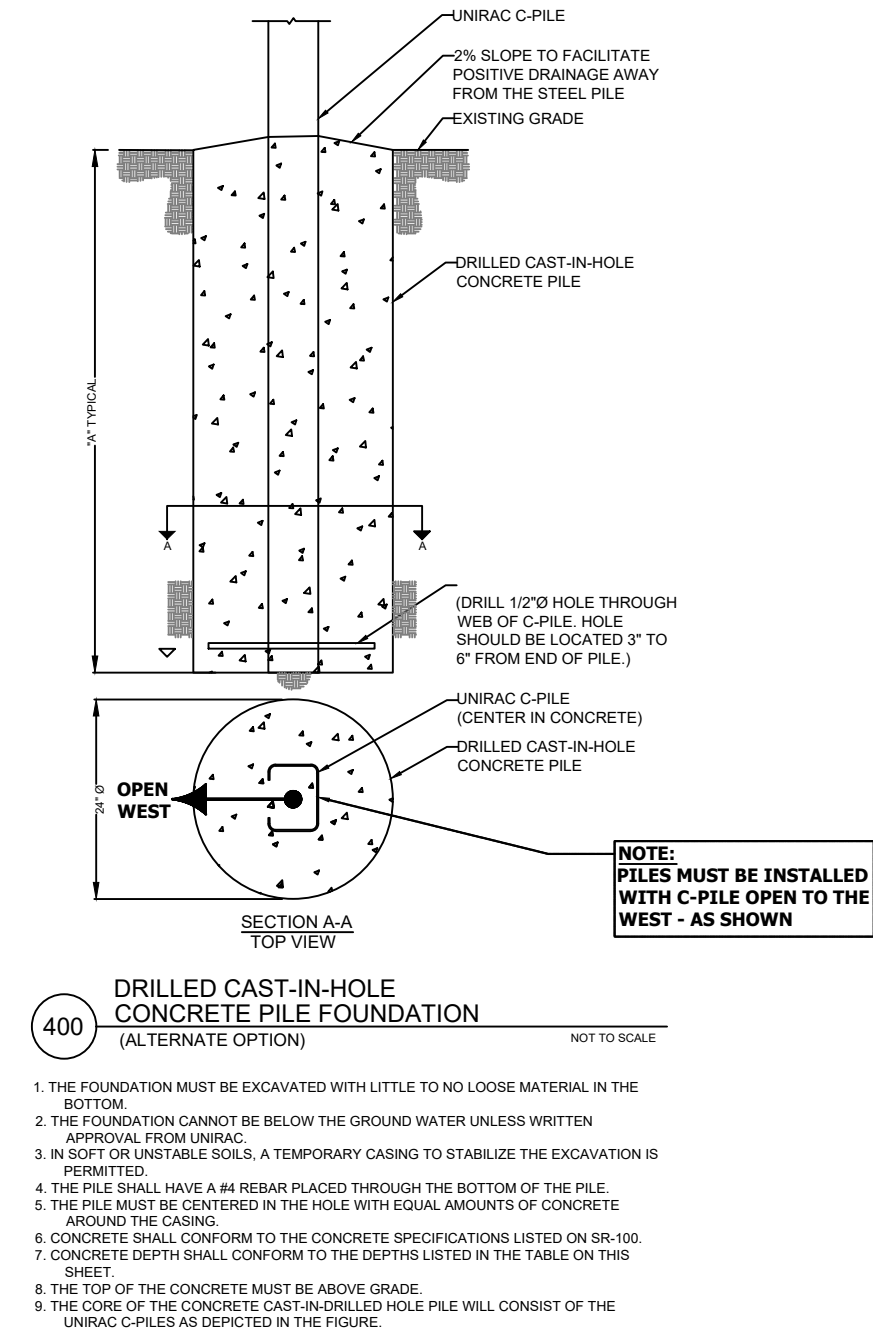
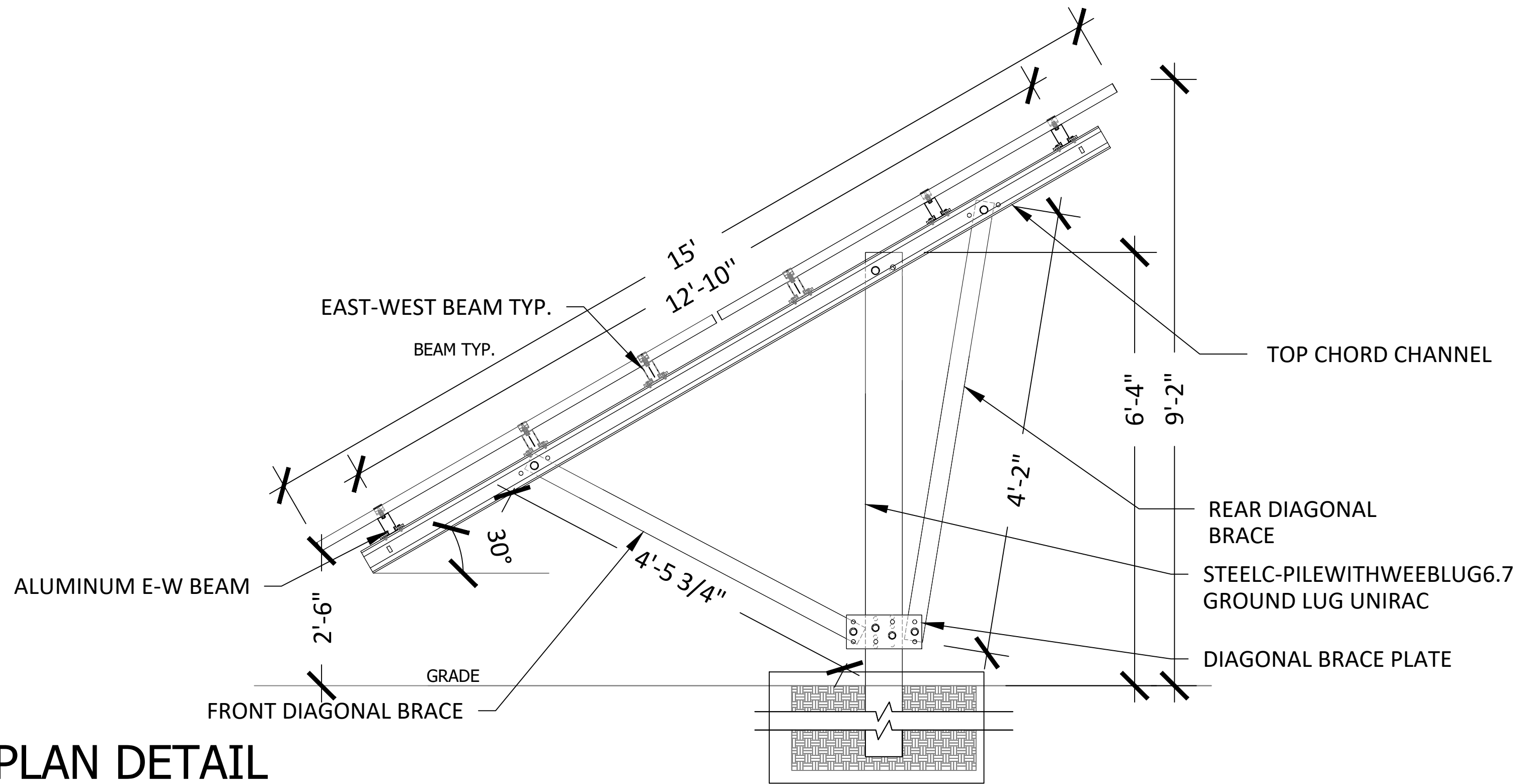
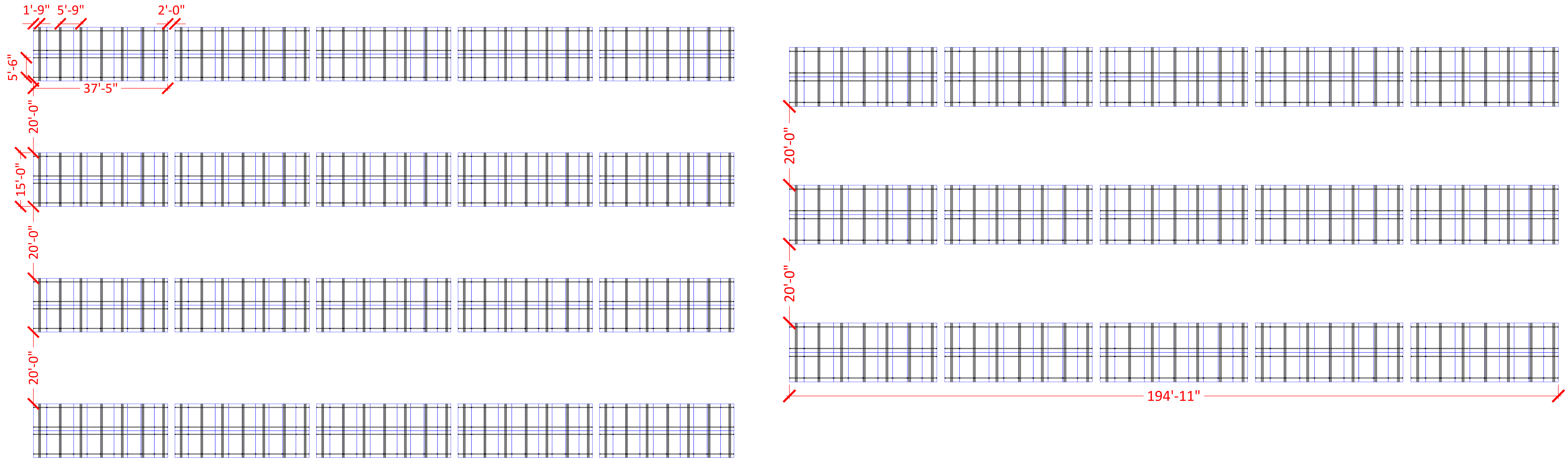
SIGNED & STAMP BY

PV-3

SHEET NAME

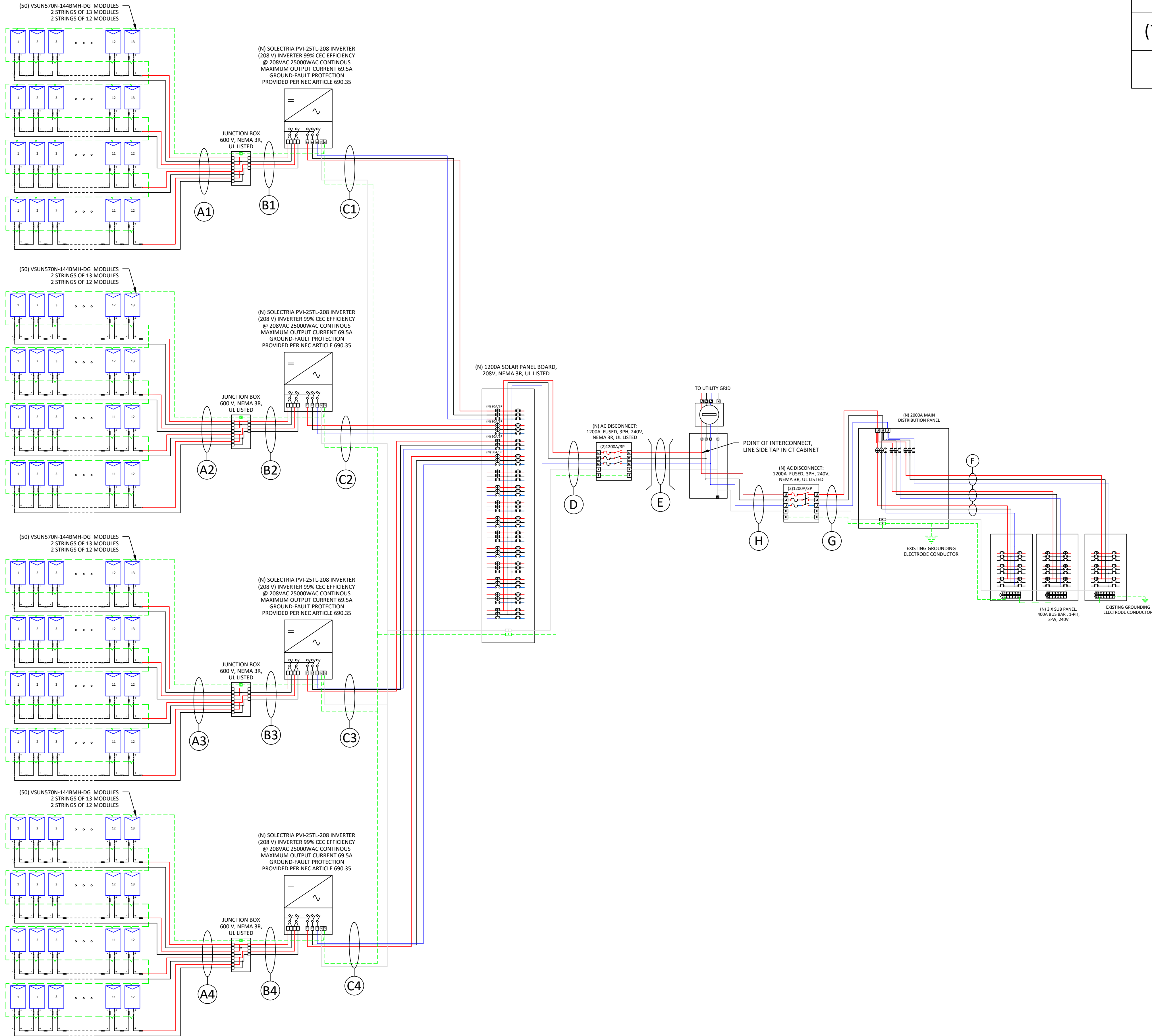
BOM &  
STRING  
LAYOUT

SHEET INFORMATION



A 4-UP - PV SYSTEM PLAN DETAIL  
Scale: 3/32"=1'-0"

|                |            |          |                                                                             |   |                 |                   |  |                                                        |
|----------------|------------|----------|-----------------------------------------------------------------------------|---|-----------------|-------------------|--|--------------------------------------------------------|
|                | DATE       |          | GROUND MOUNT COMMERCIAL<br>1234 STREET NAME, CITY, IL 9874<br><br>APN: ---- | # | REVISION/UPDATE | DATE              |  | S-1<br>SHEET NAME<br>STRUCTURAL<br>DETAILED<br>DIAGRAM |
|                | PROJECT ID |          |                                                                             |   |                 |                   |  |                                                        |
|                | CREATED BY | AK       |                                                                             |   |                 |                   |  |                                                        |
|                | CHECKED BY | MSB      |                                                                             |   |                 |                   |  |                                                        |
|                | SCALE      | NTS      |                                                                             |   |                 |                   |  |                                                        |
|                | SHEET SIZE | 24"X 36" |                                                                             |   |                 |                   |  |                                                        |
|                |            | ANSI D   |                                                                             |   |                 |                   |  |                                                        |
| HOMEOWNER INFO |            |          | SIGNED & STAMP BY                                                           |   |                 | SHEET INFORMATION |  |                                                        |



| [ DC ] SYSTEM SIZE   | [ AC ] SYSTEM SIZE      |
|----------------------|-------------------------|
| (700) MODULES x 575W | (14) INVERTERS x 15000W |
| 402.500 kW           | 195.00 kW               |

|            |          |
|------------|----------|
| DATE       |          |
| PROJECT ID |          |
| CREATED BY | AK       |
| CHECKED BY | MSB      |
| SCALE      | NTS      |
| SHEET SIZE | 24"X 36" |
|            | ANSI D   |

GROUND MOUNT COMMERICAL

1234 STREET NAME, CITY, IL 9874

APN: ----

HOMEOWNER INFO

| # | REVISION/UPDATE | DATE |
|---|-----------------|------|
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|   |                 |      |
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SIGNED & STAMP BY

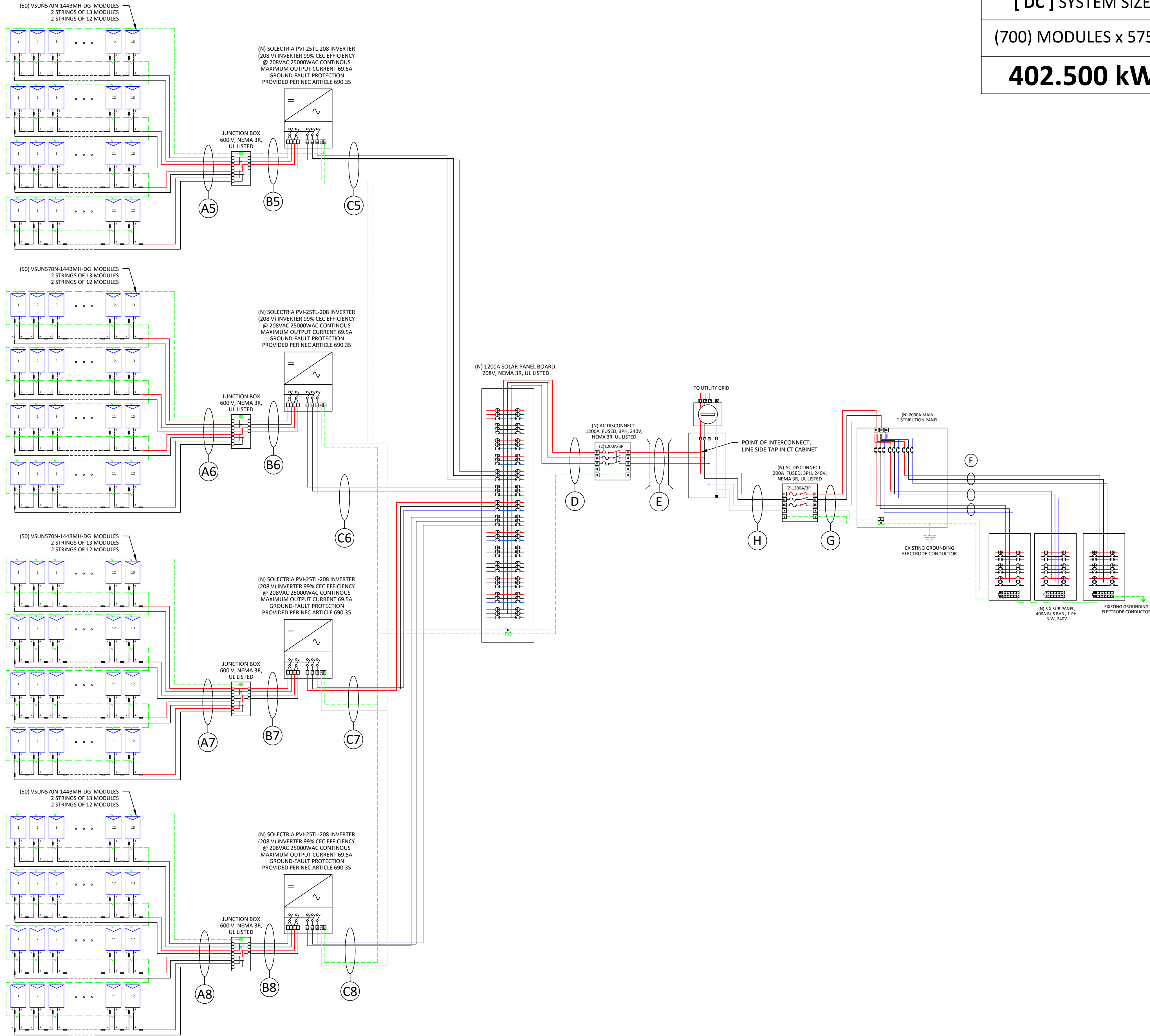
E-1

SHEET NAME

ELECTRICAL  
DIAGRAM

SHEET INFORMATION

| [ DC ] SYSTEM SIZE   | [ AC ] SYSTEM SIZE      |
|----------------------|-------------------------|
| (700) MODULES x 575W | (14) INVERTERS x 15000W |
| 402.500 kW           | 195.00 kW               |



|            |          |
|------------|----------|
| DATE       |          |
| PROJECT ID |          |
| CREATED BY | AK       |
| CHECKED BY | MSB      |
| SCALE      | NTS      |
| SHEET SIZE | 24"X 36" |
|            | ANSI D   |

## GROUND MOUNT COMMERICAL

1234 STREET NAME, CITY, IL 9874

APN: ----

HOMEOWNER INFO

| # | REVISION/UPDATE | DATE |
|---|-----------------|------|
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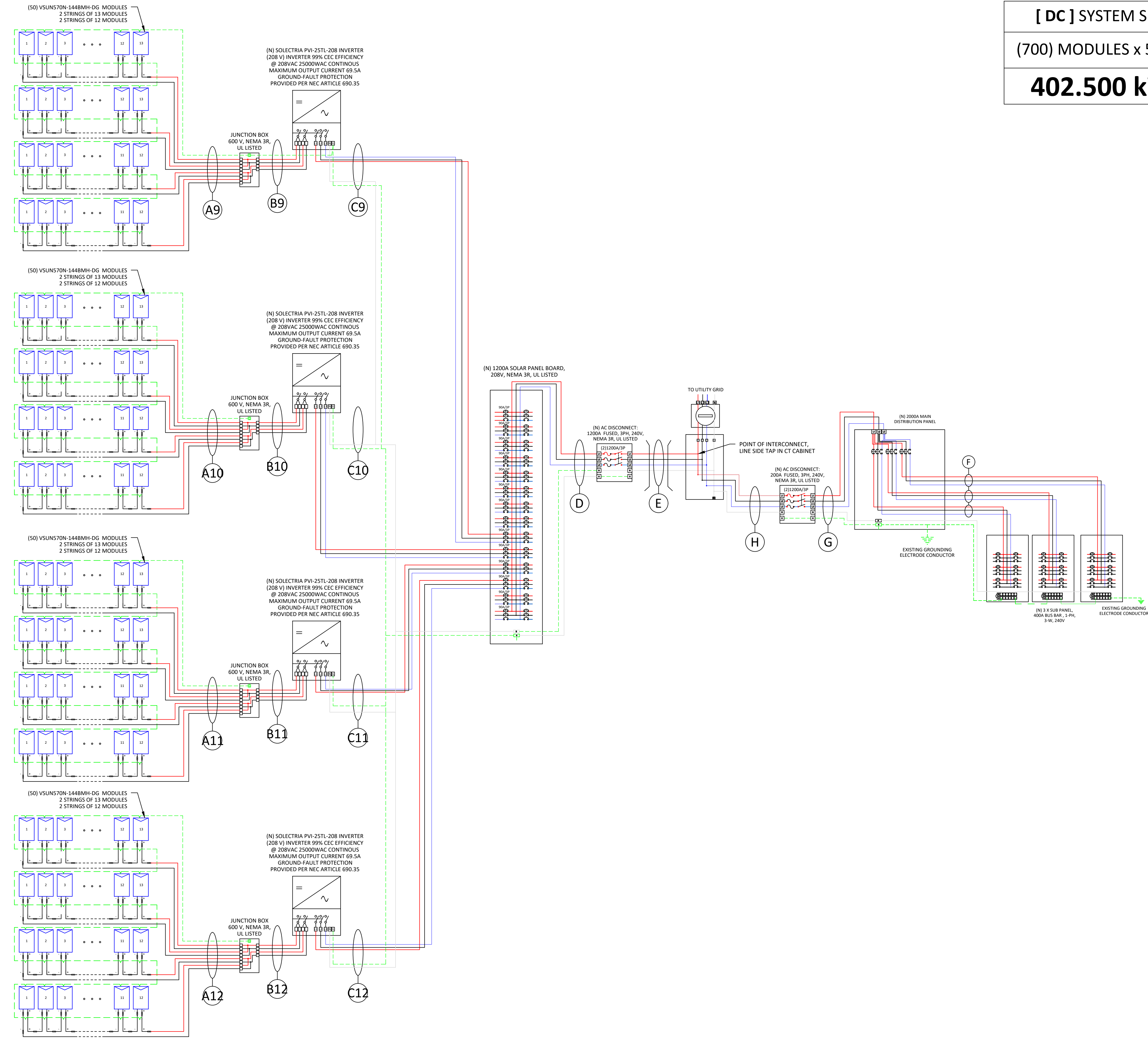
SIGNED & STAMP BY

E-2

SHEET NAME

ELECTRICAL  
DIAGRAM

SHEET INFORMATION



| [ DC ] SYSTEM SIZE   | [ AC ] SYSTEM SIZE      |
|----------------------|-------------------------|
| (700) MODULES x 575W | (14) INVERTERS x 15000W |
| 402.500 kW           | 195.00 kW               |

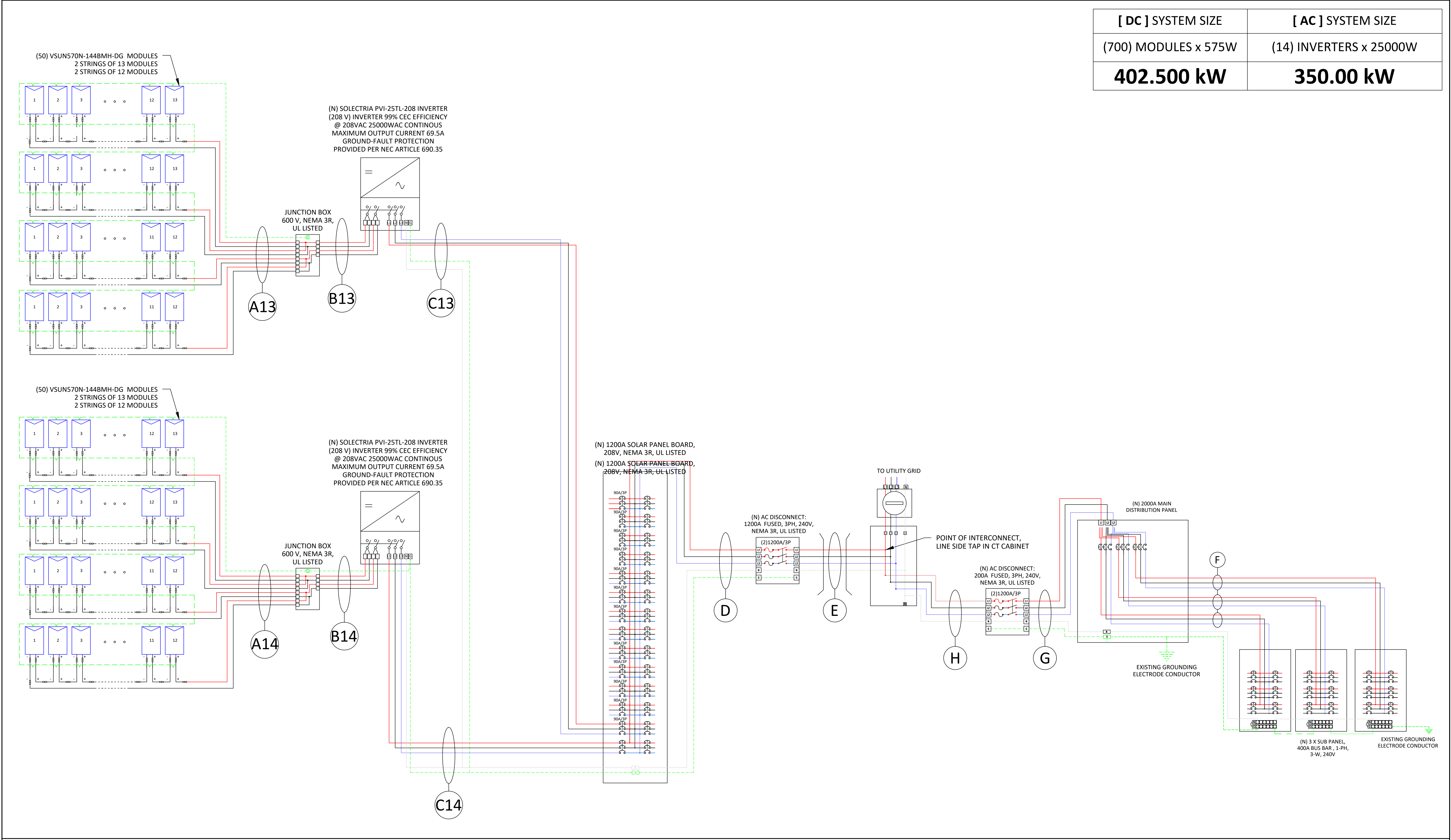
|            |          |
|------------|----------|
| DATE       |          |
| PROJECT ID |          |
| CREATED BY | AK       |
| CHECKED BY | MSB      |
| SCALE      | NTS      |
| SHEET SIZE | 24"X 36" |
|            | ANSI D   |

|                                                                                                    |                           |
|----------------------------------------------------------------------------------------------------|---------------------------|
| <div>GROUND MOUNT COMMERICAL</div> <div>1234 STREET NAME, CITY, IL 9874</div> <div>APN: ----</div> | <div>HOMEOWNER INFO</div> |
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| # | REVISION/UPDATE | DATE |
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| SIGNED & STAMP BY |

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|-----------------------|
| E-3                   |
| SHEET NAME            |
| ELECTRICAL<br>DIAGRAM |
| SHEET INFORMATION     |



| [ DC ] SYSTEM SIZE   | [ AC ] SYSTEM SIZE      |
|----------------------|-------------------------|
| (700) MODULES x 575W | (14) INVERTERS x 25000W |
| 402.500 kW           | 350.00 kW               |

|            |            |                |                                                                         |   |                 |      |  |                   |                                            |                   |
|------------|------------|----------------|-------------------------------------------------------------------------|---|-----------------|------|--|-------------------|--------------------------------------------|-------------------|
|            | DATE       |                | GROUND MOUNT COMMERCIAL<br>1234 STREET NAME, CITY, IL 9874<br>APN: ---- | # | REVISION/UPDATE | DATE |  | SIGNED & STAMP BY | E-4<br>SHEET NAME<br>ELECTRICAL<br>DIAGRAM | SHEET INFORMATION |
|            | PROJECT ID |                |                                                                         |   |                 |      |  |                   |                                            |                   |
|            | CREATED BY | AK             |                                                                         |   |                 |      |  |                   |                                            |                   |
|            | CHECKED BY | MSB            |                                                                         |   |                 |      |  |                   |                                            |                   |
|            | SCALE      | NTS            |                                                                         |   |                 |      |  |                   |                                            |                   |
| SHEET SIZE | 24"X 36"   | HOMEOWNER INFO |                                                                         |   |                 |      |  |                   |                                            |                   |
|            | ANSI D     |                |                                                                         |   |                 |      |  |                   |                                            |                   |

| [ DC ] SYSTEM SIZE   | [ AC ] SYSTEM SIZE      |
|----------------------|-------------------------|
| (700) MODULES x 575W | (14) INVERTERS x 25000W |
| 402.500 kW           | 350.00 kW               |

| INVERTER SPECIFICATIONS   |                                         |
|---------------------------|-----------------------------------------|
| MANUFACTURER / MODEL      | SOLECTRIA PVI-25TL-208 INVERTER (208 V) |
| NOMINAL OUTPUT VOLTAGE    | 240VAC                                  |
| CONTINUOUS OUTPUT CURRENT | 69.5A                                   |

| SOLAR MODULE SPECIFICATIONS |                       |
|-----------------------------|-----------------------|
| MANUFACTURER / MODEL        | VSUN570N-144BMH-DG    |
| VMP                         | 42.53V                |
| IMP                         | 13.54A                |
| VOC                         | 51.26V                |
| ISC                         | 14.31A                |
| TEMP. COEFF. VOC            | -0.26 %/°C            |
| MODULE DIMENSION            | 89.68 (L) x 44.64 (W) |
| PANEL WATTAGE               | 575W                  |

| AMBIENT TEMPERATURE SPECS             |            |
|---------------------------------------|------------|
| RECORD LOW TEMP                       | -19°C      |
| AMBIENT TEMP (HIGH TEMP 2%)           | 35°C       |
| CONDUIT HEIGHT                        | 7/8"       |
| ROOF TOP TEMP                         | 90°C       |
| CONDUCTOR TEMPERATURE RATE            | 57°C       |
| MODULE TEMPERATURE COEFFICIENT OF VOC | -0.26 %/°C |

| EQUIPMENT DETAILS        |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| (#) PV MODULES           | (700) VSUN570N-144BMH-DG                                            |
| (#) MICROINVERTERS       | (14) SOLECTRIA PVI-25TL-208 INVERTER (208 V)                        |
| OUTPUT CURRENT           | 973.0A                                                              |
| STRINGS                  | 28 STRINGS OF 13 MODULES & 28 STRINGS OF 12 MODULES                 |
| POINT OF INTERCONNECTION | LINE SIDE TAP IN THE CT CABINET , 1200A BUSBAR AND 120/240V, 3W, 1φ |
| UTILITY                  | ----                                                                |

|            |          |
|------------|----------|
| DATE       |          |
| PROJECT ID |          |
| CREATED BY | AK       |
| CHECKED BY | MSB      |
| SCALE      | NTS      |
| SHEET SIZE | 24"X 36" |
|            | ANSI D   |

GROUND MOUNT COMMERICAL

1234 STREET NAME, CITY, IL 9874

APN: ----

HOMEOWNER INFO

| # | REVISION/UPDATE | DATE |
|---|-----------------|------|
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SIGNED & STAMP BY

E-5

SHEET NAME

ELECTRICAL  
DIAGRAM

SHEET INFORMATION

| ID   | CONDUCTOR  |         |        | CONDUIT           | # OF<br>PARALLEL<br>CIRCUITS | CURRENT-CARRYING<br>CONDUCTORS IN<br>CONDUIT | CONDUIT FILL<br>PERCENT | OCPD   | EGC    |                | TEMP. CORR.<br>FACTOR |        | CONDUIT<br>FILL FACTOR | CONT. CURRENT |   |      |   | MAX<br>CURRENT | BASE<br>AMP. | DERATED AMP.              | LENGTH | VOLTAGE<br>DROP |
|------|------------|---------|--------|-------------------|------------------------------|----------------------------------------------|-------------------------|--------|--------|----------------|-----------------------|--------|------------------------|---------------|---|------|---|----------------|--------------|---------------------------|--------|-----------------|
| A.1  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.2  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.3  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.4  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.5  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.6  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.7  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.8  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.9  | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.10 | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.11 | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.12 | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.13 | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| A.14 | 10 AWG     | PV WIRE |        | -                 | 1                            | 2                                            | N/A                     | N/A    | 6 AWG  | BARE COPPER    | 0.71                  | (57°C) | N/A                    | 14.3A         | x | 1.25 | = | 17.9A          | N/A          | N/A                       | 195FT  | 0.94%           |
| B.1  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 5FT    | 0.11%           |
| B.2  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 12FT   | 0.26%           |
| B.3  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 5FT    | 0.11%           |
| B.4  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 12FT   | 0.26%           |
| B.5  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 5FT    | 0.11%           |
| B.6  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 12FT   | 0.26%           |
| B.7  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 5FT    | 0.11%           |
| B.8  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 12FT   | 0.26%           |
| B.9  | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 5FT    | 0.11%           |
| B.10 | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 12FT   | 0.26%           |
| B.11 | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 5FT    | 0.11%           |
| B.12 | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 12FT   | 0.26%           |
| B.13 | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 5FT    | 0.11%           |
| B.14 | 8 AWG      | THWN-2  | COPPER | MIN 0.75" Dia EMT | 2                            | 4                                            | 33.41%                  | N/A    | 8 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 0.8                    | 28.6A         | x | 1.25 | = | 35.8A          | 55A          | 55A x 0.8 x 0.96 = 42.2A  | 12FT   | 0.26%           |
| C.1  | 3 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 225FT  | 6.53%           |
| C.2  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 225FT  | 6.53%           |
| C.3  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 190FT  | 5.51%           |
| C.4  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 190FT  | 5.51%           |
| C.5  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 155FT  | 4.50%           |
| C.6  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 155FT  | 4.50%           |
| C.7  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 120FT  | 3.48%           |
| C.8  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 120FT  | 3.48%           |
| C.9  | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 85FT   | 2.47%           |
| C.10 | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 85FT   | 2.47%           |
| C.11 | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 50FT   | 1.45%           |
| C.12 | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 50FT   | 1.45%           |
| C.13 | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 15FT   | 0.44%           |
| C.14 | 4 AWG      | THWN-2  | COPPER | MIN 1.25" Dia PVC | 1                            | 4                                            | 32.36%                  | 90A    | 6 AWG  | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 69.5A         | x | 1.25 | = | 86.9A          | 115A         | 115A x 0.8 x 0.96 =88.3A  | 15FT   | 0.44%           |
| D    | 1250 kcmil | THWN-2  | COPPER | MIN 5" Dia PVC    | 4                            | 16                                           | 32.31%                  | 12000A | 4/0AWG | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 243.3A        | x | 1.25 | = | 304.1A         | 665A         | 430A x 0.5 x 0.96 =319.2A | 5FT    | 0.66%           |
| E    | 1250 kcmil | THWN-2  | COPPER | MIN 5" Dia PVC    | 4                            | 16                                           | 32.31%                  | N/A    | -      | -              | 0.96                  | (35°C) | 1                      | 243.3A        | x | 1.25 | = | 304.1A         | 665A         | 430A x 0.5 x 0.96 =319.2A | 142FT  | 0.02%           |
| F    | 1000 kcmil | THWN-2  | COPPER | MIN 5" Dia PVC    | 3                            | 9                                            | 32.31%                  | N/A    | 4/0AWG | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 320.0A        | x | 1.25 | = | 400.0A         | 615A         | 615A x 0.7 x 0.96 =413.3A | 5FT    | 0.02%           |
| G    | 1250 kcmil | THWN-2  | COPPER | MIN 5" Dia PVC    | 4                            | 16                                           | 32.31%                  | 12000A | 4/0AWG | THWN-2, COPPER | 0.96                  | (35°C) | 1                      | 240.0A        | x | 1.25 | = | 300.0A         | 665A         | 430A x 0.5 x 0.96 =319.2A | 5FT    | 0.01%           |
| H    | 1250 kcmil | THWN-2  | COPPER | MIN 5" Dia PVC    | 4                            | 16                                           | 32.31%                  | N/A    | -      | -              | 0.96                  | (35°C) | 1                      | 240.0A        | x | 1.25 | = | 300.0A         | 665A         | 430A x 0.5 x 0.96 =319.2A | 5FT    | 0.01%           |

|                |            |          |                                                                                                    |   |                 |      |                                  |                                                                                                      |
|----------------|------------|----------|----------------------------------------------------------------------------------------------------|---|-----------------|------|----------------------------------|------------------------------------------------------------------------------------------------------|
|                | DATE       |          | <div>GROUND MOUNT COMMERICAL</div> <div>1234 STREET NAME, CITY, IL 9874</div> <div>APN: ----</div> | # | REVISION/UPDATE | DATE | <div>SIGNED &amp; STAMP BY</div> | <div>E-6</div> <div>SHEET NAME</div> <div>ELECTRICAL CALCULATIONS</div> <div>SHEET INFORMATION</div> |
|                | PROJECT ID |          |                                                                                                    |   |                 |      |                                  |                                                                                                      |
|                | CREATED BY | AK       |                                                                                                    |   |                 |      |                                  |                                                                                                      |
|                | CHECKED BY | MSB      |                                                                                                    |   |                 |      |                                  |                                                                                                      |
|                | SCALE      | NTS      |                                                                                                    |   |                 |      |                                  |                                                                                                      |
|                | SHEET SIZE | 24"X 36" |                                                                                                    |   |                 |      |                                  |                                                                                                      |
|                |            | ANSI D   |                                                                                                    |   |                 |      |                                  |                                                                                                      |
| HOMEOWNER INFO |            |          |                                                                                                    |   |                 |      |                                  |                                                                                                      |

! WARNING

ELECTRIC SHOCK HAZARD

DO NOT TOUCH TERMINALS.  
TERMINALS ON BOTH LINE AND LOAD SIDES MAY  
BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION:  
MAIN SERVICE PANEL/AC DISCONNECT/INVERTER/  
AC COMBINER  
2022 CEC 690.13(B)

! WARNING

ELECTRIC SHOCK HAZARD

IF GROUND FAULT IS INDICATED  
ALL NORMALLY GROUNDED  
CONDUCTORS MAY BE  
UNGROUND AND ENERGIZED

LABEL LOCATION:  
INVERTER(S), ENPHASE ENVOY ENCLOSURE (IF  
APPLICABLE).  
PER CODE(S): CEC 2022: 690.5(C)

PHOTOVOLTAIC AC DISCONNECT

RATED AC OUTPUT CURRENT  973.00 A

NOMINAL OPERATING AC VOLTAGE  240 VAC

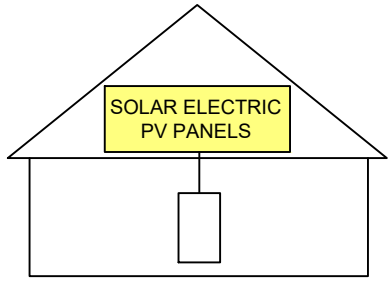
LABEL LOCATION:  
MAIN SERVICE PANEL/AC DISCONNECT  
2022 CEC 690. 13(B)

RAPID SHUTDOWN SWITCH  
FOR SOLAR PV SYSTEM

LABEL LOCATION:  
INVERTER  
AT OR WITHIN 3' OF THE DC COMBINER  
SWITCH  
2022 CEC 690.56(C)(3)

SOLAR PV SYSTEM EQUIPPED  
WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN  
SWITCH TO THE  
"OFF" POSITION TO  
SHUT DOWN PV SYSTEM  
AND REDUCE  
SHOCK HAZARD  
IN THE ARRAY



LABEL LOCATION:  
MAIN SERVICE PANEL  
IF MSD IS OUTSIDE PLACE IT THERE / IF  
MSD IS INSIDE PLACE ON THE AC DISCONNECT  
2022 CEC 690.56(C)(1)(a)

PHOTOVOLTAIC SYSTEM  
UTILITY DISCONNECT SWITCH

LABEL LOCATION:  
AC DISCONNECT  
2022 CEC 690.56(C)(3)

! CAUTION

DUAL POWER SOURCE  
SECOND SOURCE IS  
PHOTOVOLTAIC

LABEL LOCATION:  
MAIN SERVICE PANEL/AC DISCONNECT/AC COMBINER/  
REVENUE METER  
2022 CEC 705.12(B)(3)

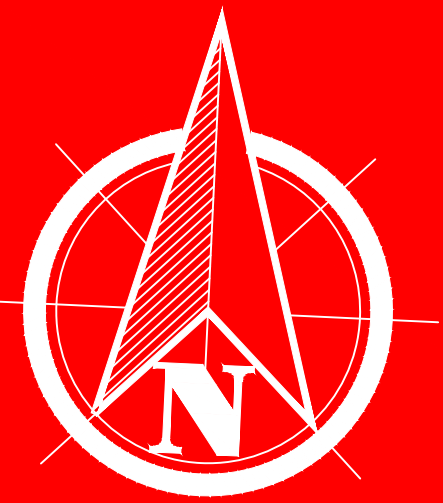
PHOTOVOLTAIC SYSTEM  
COMBINER PANEL  
DO NOT ADD LOADS

LABEL LOCATION:  
PHOTOVOLTAIC AC COMBINER (IF APPICABLE).  
PER CODE(S): CEC 2020: 705.12(B)(2)(3)(c)

ADHESIVE FASTENED SIGNS:

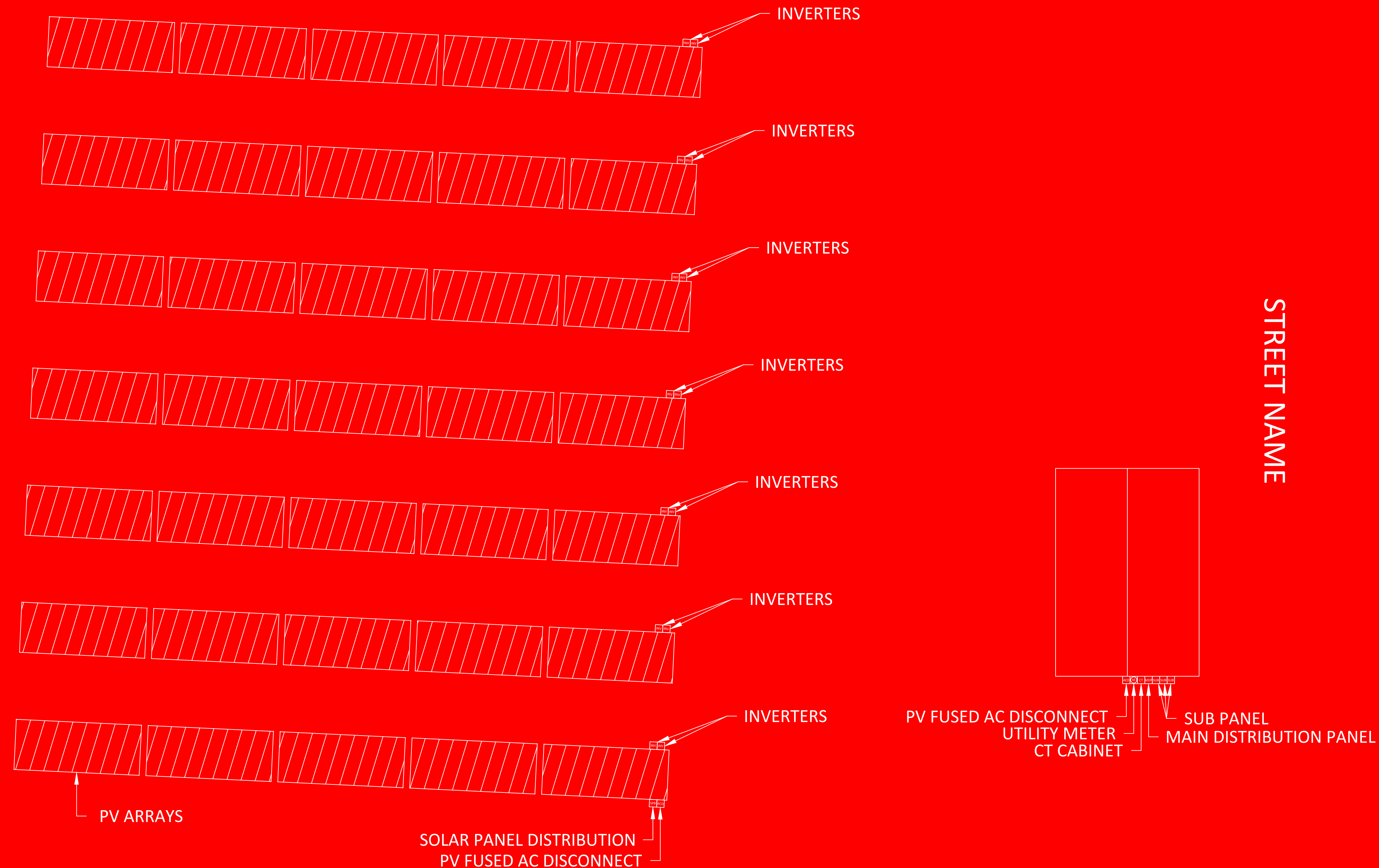
- LABELS SHALL BE MADE OF RED PLASTIC MATERIAL WITH ENGRAVED WHITE LETTERS.
- LETTERS SHALL BE A MINIMUM 3/8" IN SIZE.
- THE LABELS SHALL BE PERMANENTLY ATTACHED TO THE APPROPRIATE PANEL.
- AC & DC CONDUIT, RACEWAY, ENCLOSURES, CABLE ASSEMBLIES AND JUNCTION BOXES SHALL BE RED BACKGROUND MATERIAL WITH WHITE LETTERING MADE OF DURABLE ADHESIVE, REFLECTIVE, WEATHER RESISTANT MATERIAL SUITABLE FOR THE ENVIRONMENT PER UL 969; TO ALERT FIRE SERVICE TO AVOID CUTTING THEM OFF.

|                |            |          |                                                                             |   |                 |      |                                  |                   |
|----------------|------------|----------|-----------------------------------------------------------------------------|---|-----------------|------|----------------------------------|-------------------|
|                | DATE       |          | GROUND MOUNT COMMERICAL<br>1234 STREET NAME, CITY, IL 9874<br><br>APN: ---- | # | REVISION/UPDATE | DATE | <div>SIGNED &amp; STAMP BY</div> | E-7<br>SHEET NAME |
|                | PROJECT ID |          |                                                                             |   |                 |      |                                  |                   |
|                | CREATED BY | AK       |                                                                             |   |                 |      |                                  |                   |
|                | CHECKED BY | MSB      |                                                                             |   |                 |      |                                  |                   |
|                | SCALE      | NTS      |                                                                             |   |                 |      |                                  |                   |
|                | SHEET SIZE | 24"X 36" |                                                                             |   |                 |      |                                  |                   |
|                |            | ANSI D   |                                                                             |   |                 |      |                                  |                   |
| HOMEOWNER INFO |            |          | SHEET INFORMATION                                                           |   |                 |      |                                  |                   |



# CAUTION!

POWER TO THIS BUILDING IS ALSO SUPPLIED FROM THE FOLLOWING SOURCES  
WITH DISCONNECTS LOCATED AS SHOWN.



LABEL LOCATION:  
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER  
PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

|            |          |
|------------|----------|
| DATE       |          |
| PROJECT ID |          |
| CREATED BY | AK       |
| CHECKED BY | MSB      |
| SCALE      | NTS      |
| SHEET SIZE | 24"X 36" |
|            | ANSI D   |

GROUND MOUNT COMMERICAL

1234 STREET NAME, CITY, IL 9874

APN: ----

HOMEOWNER INFO

| # | REVISION/UPDATE | DATE |
|---|-----------------|------|
|   |                 |      |
|   |                 |      |
|   |                 |      |
|   |                 |      |
|   |                 |      |
|   |                 |      |

SIGNED & STAMP BY

E-8

SHEET NAME

SHEET INFORMATION



**VSUN**  
Innovative & Smart



## VSUN575N-144BMH-DG

VSUN570N-144BMH-DG VSUN565N-144BMH-DG  
VSUN560N-144BMH-DG

**430W**  
Highest power output

**22.02%**  
Module efficiency

**1.0%**  
First-year degradation warranty

**0.4%**  
Annual degradation over 30 years

### KEY FEATURES

- N-type TOPCon Technology
- Higher output power
- MBB technology with Circular Ribbon
- Positive tolerance offer
- Lower LID
- Better shading tolerance
- Better temperature coefficient
- Excellent PID Resistance
- Lower LCOE

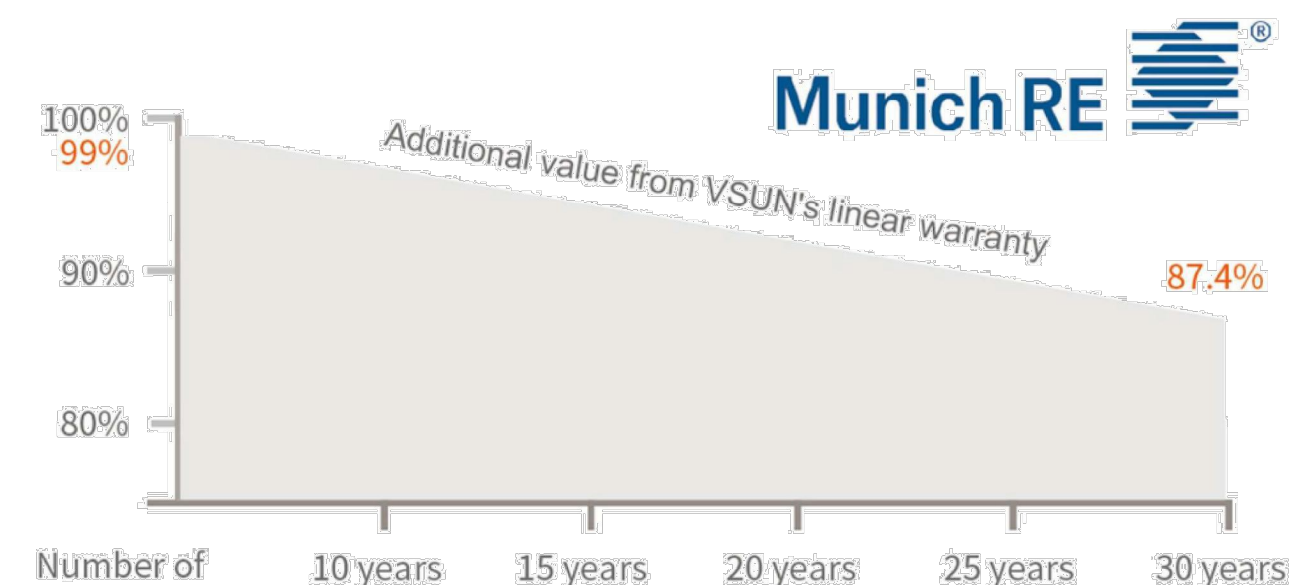
### ABOUT VSUN

Invested by Fuji Solar, VSUN SOLAR is a solar solution provider with headquartered in Tokyo, Japan that offers reliability, high efficiency solar products and technology globally. VSUN is rated as BNEF Tier 1 PV module manufacturer, PVEL Lab "Best performer" and EcoVadis "Bronze Award".

### PRODUCT CERTIFICATION



### PRODUCT CERTIFICATION



### Electrical Characteristics at Standard Test Conditions(STC)

| Module Type                      | VSUN575N-144BMH-DG | VSUN570N-144BMH-DG | VSUN565N-144BMH-DG | VSUN560N-144BMH-DG |
|----------------------------------|--------------------|--------------------|--------------------|--------------------|
| Maximum Power - Pmax (W)         | 575                | 570                | 565                | 560                |
| Open Circuit Voltage - Voc (V)   | 51.26              | 51.07              | 50.87              | 50.67              |
| Short Circuit Current - Isc (A)  | 14.31              | 14.25              | 14.19              | 14.13              |
| Maximum Power Voltage - Vmpp (V) | 42.53              | 42.34              | 42.14              | 41.95              |
| Maximum Power Current - Impp (A) | 13.54              | 13.48              | 13.41              | 13.34              |
| Module Efficiency                | 22.26%             | 22.07%             | 21.87%             | 21.68%             |

Standard Test Conditions (STC): irradiance 1,000 W/m<sup>2</sup>; AM 1.5; module temperature 25°C. Pmax Sorting : 0~5W. Measuring Tolerance: ±3%.

Remark: Electrical data do not refer to a single module and they are not part of the offer. They only serve for comparison among different module types.

### Electrical Characteristics with different rear side power gain(reference to 570 front)

| Pmax (W) | Voc (V) | Isc (A) | Vmpp (V) | Impp (A) | Pmax gain |
|----------|---------|---------|----------|----------|-----------|
| 599      | 51.07   | 14.96   | 42.34    | 14.15    | 5%        |
| 628      | 51.07   | 15.68   | 42.34    | 14.83    | 10%       |
| 684      | 51.11   | 17.10   | 42.29    | 16.18    | 20%       |
| 713      | 51.11   | 17.81   | 42.29    | 16.85    | 25%       |

### Material Characteristics

|                    |                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------|
| Dimensions         | 2278×1134×35mm (L×W×H)                                                                       |
| Weight             | 32.7kg                                                                                       |
| Frame              | Silver anodized aluminum profile                                                             |
| Front Glass        | AR-coating Semi-toughened glass, 2.0mm                                                       |
| Cell Encapsulation | EVA (Ethylene-Vinyl-Acetate) or POE                                                          |
| Back Glass         | Glazed & Semi-toughened glass, 2.0mm                                                         |
| Cells              | 12×12 pieces bifacial monocrystalline solar cells series strings                             |
| Junction Box       | IP68, 3 diodes                                                                               |
| Cable&Connector    | Potrait: 500 mm (cable length can be customized) , 1×4 mm <sup>2</sup> , compatible with MC4 |

### Packaging

|                   |                   |
|-------------------|-------------------|
| Dimensions(L×W×H) | 2310×1125×1253mm  |
| Container 20'     | 150               |
| Container 40'     | 300               |
| Container 40'HC   | 600 or 540 for US |

### Temperature Characteristics

|                                 |            |
|---------------------------------|------------|
| NOCT                            | 45°C(±2°C) |
| Voltage Temperature Coefficient | -0.26%/°C  |
| Current Temperature Coefficient | +0.046%/°C |
| Power Temperature Coefficient   | -0.30%/°C  |

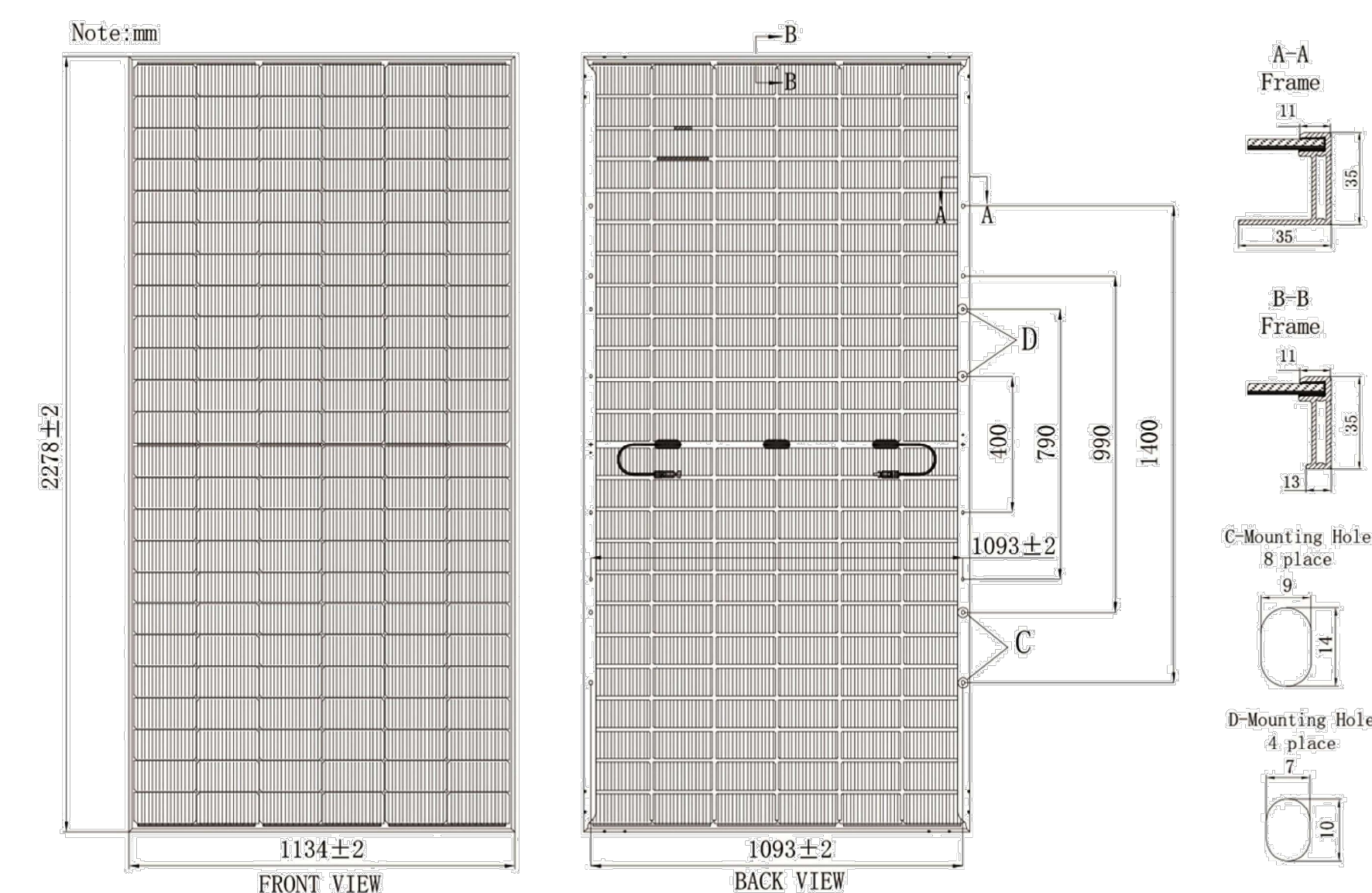
### Maximum Ratings

|                            |        |
|----------------------------|--------|
| Maximum System Voltage [V] | 1500   |
| Series Fuse Rating [A]     | 30     |
| Bifaciality                | 80%±5% |

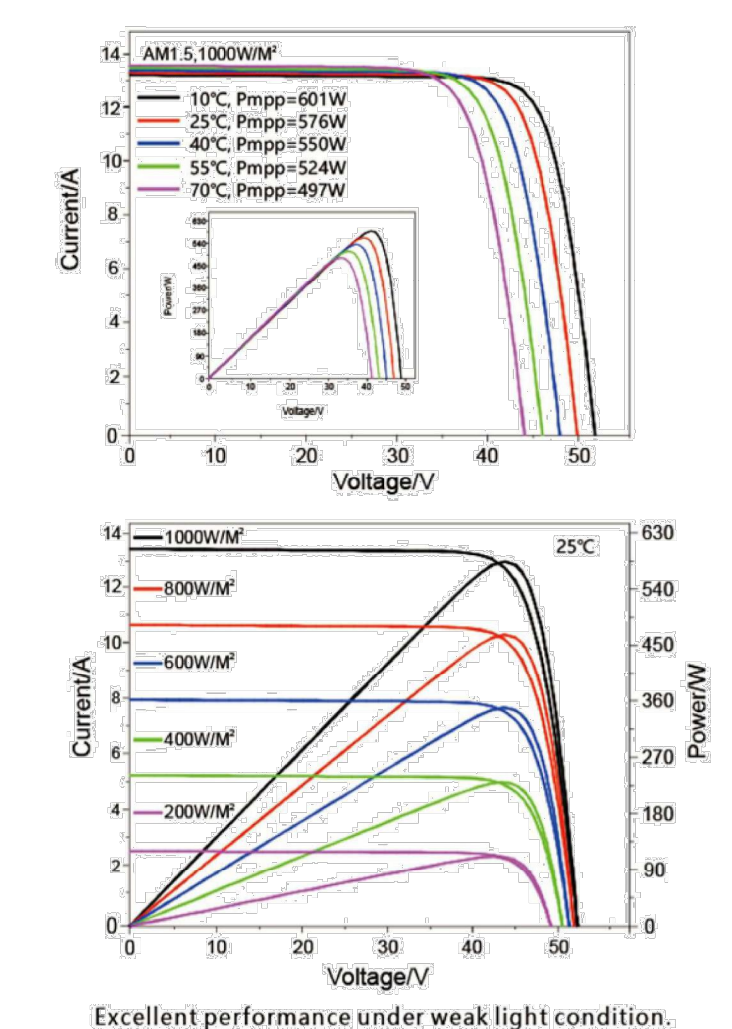
### System Design

|                      |                                                       |
|----------------------|-------------------------------------------------------|
| Temperature Range    | -40 °C to +85 °C                                      |
| Withstanding Hail    | Maximum diameter of 25 mm with impact speed of 23 m/s |
| Maximum Surface Load | 5,400 Pa                                              |
| Application class    | class A                                               |

### Dimensions



### IV-Curves

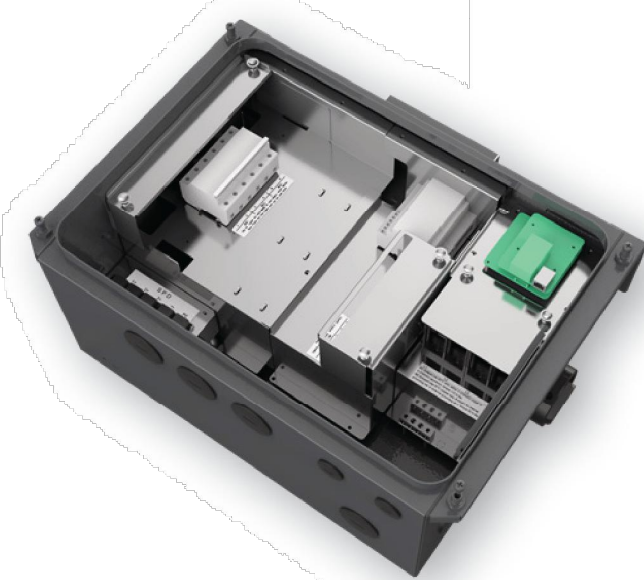


# SOLECTRIA® PVI-25TL-208

25 KW, 208 VAC, 1000 VDC STRING INVERTERS

### Features

- Certified to IEEE 1547-2018 and UL 1741SB
- Wirebox models with built-in SunSpec compliant transmitters for Module-Level Rapid Shutdown for simple, safe NEC compliance
- UL Listed as a PV Rapid Shutdown System with APsmart
- Integrated UL-listed Arc-Fault protection
- 15-90° mounting angle allows low-profile rooftop installations
- 3 MPPT's with 2 inputs each for PV array flexibility
- Industry-leading DC/AC ratio of 1.8
- Integrated AC and DC disconnects
- Remote firmware upgrades and diagnostics
- NEMA 4X outdoor-rated enclosure with proven performance
- Copper and Aluminum compatible AC connections
- Compatible with Bifacial PV Modules



Rapid Shutdown Ready Wirebox

Yaskawa Solectria Solar's PVI-25TL-208 25kW (25kVA) three phase string inverters are designed for rooftop and carport applications



### PVI-25TL-208 DESIGN

These high performance, advanced and reliable inverters are designed specifically for the North American environment and grid.

High efficiency at 97.0% peak and 96.5% CEC, wide operating voltages, broad temperature ranges and a NEMA Type 4X enclosure enable this inverter platform to operate at high performance across many applications.

The product ships with the Rapid Shutdown Ready wirebox, fully integrated and separable with touch-safe fusing, monitoring, and AC and DC disconnect switches.

The integrated Sunspec compliant PLC transmitter in the wirebox enables PVRSS certified module-level rapid shutdown when used with APsmart products.

The Ethernet Network Card enables monitoring, controls and remote product upgrades.



Yaskawa Solectria Solar 1-978-683-9700 | Email: sales@solectria.com | solectria.com  
Document No. FL.PVI25TL-208.01 | 07/17/2024 | © 2021 Yaskawa America, Inc.

## SOLECTRIA® PVI-25TL-208 TECHNICAL DATA

### SPECIFICATIONS

| PVI-25TL-208 Commercial Transformerless String Inverter<br>(Listing File Model: PVI 25TL-208) |                                                       |                                                                                                                 |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| DC Input                                                                                      | Maximum PV Power                                      | 45 kW (17 kW per MPPT)                                                                                          |
|                                                                                               | Maximum Input Voltage                                 | 1000 VDC                                                                                                        |
|                                                                                               | DC Voltage Ranges: Operating / Maximum Power (MPPT)   | 200 - 950 VDC / 480 - 850 VDC                                                                                   |
|                                                                                               | Start-up DC Input Voltage / Power                     | 330 V / 80 W                                                                                                    |
|                                                                                               | Number of MPPT Trackers/Inputs                        | 3 Trackers / 2 Fused-Inputs each                                                                                |
|                                                                                               | Maximum Available PV Current (Isc x 1.25)             | 135 A (45 A per MPPT)                                                                                           |
|                                                                                               | DC Surge Protection                                   | Type II MOV, 2800 V <sub>CE</sub> , 20 kA <sub>ITM</sub> (8/20 μs)                                              |
| AC Output                                                                                     | Rated AC Real Power / Apparent Power / Output Current | 25 kW / 25 kVA / 69.5 A                                                                                         |
|                                                                                               | Nominal Output Voltage / Range                        | 208 VAC / -12% to +10%                                                                                          |
|                                                                                               | Nominal Output Frequency / Range                      | 60 Hz / 57-63 Hz                                                                                                |
|                                                                                               | Power Factor                                          | Unity, > 0.99 (Adjustable 0.8 leading to 0.8 lagging)                                                           |
|                                                                                               | Fault Current Contribution (1 Cycle RMS)              | 64.1 A                                                                                                          |
|                                                                                               | Total Harmonic Distortion (THD) @Rated Load           | < 3%                                                                                                            |
|                                                                                               | Grid Connection Type                                  | 3-Ph/PE/N (neutral conductor optional)                                                                          |
| Efficiency                                                                                    | Maximum OCPD Device                                   | 125 A                                                                                                           |
|                                                                                               | AC Surge Protection                                   | Type II MOV, 1240 VC, 15 kA ITM (8/20 μs)                                                                       |
|                                                                                               | Maximum Efficiency / CEC Efficiency                   | 97.0% / 96.5%                                                                                                   |
|                                                                                               | Stand-by / Night Consumption                          | < 3 W                                                                                                           |
| Environment                                                                                   | Enclosure Protection Degree                           | NEMA Type 4X                                                                                                    |
|                                                                                               | Cooling Method                                        | Variable speed cooling fans                                                                                     |
|                                                                                               | Operating Temperature Range <sup>1</sup>              | -22°F to +140°F / - 30°C to +60°C                                                                               |
|                                                                                               | Non-Operating Temperature Range                       | No low temp minimum to +158°F / +70°C maximum                                                                   |
|                                                                                               | Operating Humidity                                    | 0 to 100%                                                                                                       |
| Display and Communication                                                                     | Operating Altitude                                    | 13,123.4 ft / 4000 m (derating from 9842.5 ft / 3000 m)                                                         |
|                                                                                               | Modbus Protocol                                       | Proprietary / SunSpec                                                                                           |
|                                                                                               | SolrenView Web-based Monitoring Service               | Optional                                                                                                        |
|                                                                                               | Revenue Grade Metering                                | Optional, external                                                                                              |
|                                                                                               | Communication Interface                               | RS-485                                                                                                          |
|                                                                                               | Remote Firmware Upgrades                              | Ethernet network card required                                                                                  |
|                                                                                               | Remote Diagnostics                                    | Ethernet network card required                                                                                  |
| Safety                                                                                        | Certifications and Standards                          | IEEE 1547-2018, UL 1741-SB, UL1741-SA, UL1699B, UL1998, CSA-C22.2 NO.107.1-01, FCC Part 15 (Subpart B, Class A) |
|                                                                                               | Selectable Grid Standard                              | IEEE 1547, CA Rule 21, ISO-NE, HECO                                                                             |
|                                                                                               | Smart-Grid Features                                   | Volt-RideThru, Freq-RideThru, Ramp-Rate, Specified-PF, Volt-VAR, Freq-Watt, Volt-Watt, Watt-VAR                 |
| Warranty                                                                                      | Standard Terms                                        | 10 years                                                                                                        |
| Mechanical                                                                                    | Acoustic Noise Rating                                 | < 60 dBA @ 1m and 25°C                                                                                          |
|                                                                                               | Dimensions (H x W x D)                                | 39.4 x 23.6 x 10.24 in. (1000 x 600 x 260 mm)                                                                   |
|                                                                                               | Weight                                                | Inverter: 123.5 lbs / 56 kg; Wire-box: 33lbs / 15kg                                                             |
|                                                                                               | Mounting / Installation Angle <sup>2</sup>            | 15 to 90 degrees from horizontal (vertical or angled)                                                           |
|                                                                                               | AC Termination                                        | M8 Stud Type Terminal Block<br>(Wire range: #6 - 3/0 AWG Cu / Al, Lugs not supplied)                            |
|                                                                                               | DC Termination                                        | Screw Clamp, Neg. Busbar Wire range: #14 - #6 AWG Cu                                                            |

#### Wirebox Specifications

| Wirebox Fuse Configuration |                              | 6 Fused Positions (2 Positions per MPPT), 20A Fuses Standard (25, 30A accepted) <sup>3</sup> |                                              |
|----------------------------|------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------|
| Wirebox Versions           | APsmart Transmitter Built-In | Inverter Model: PVI-25TL-208WB-APS (only positive polarity fused)                            | MLRSD Compatibility: APsmart RSD-S and RSD-D |

1) Active Power Derating begins at 45°C when PF=1 and Vmp ≥ Vmin, and at 50°C when PF=1 and Vmp ≥ 700 Vdc.  
2) Shade Cover accessory required for installation angles of 75 degrees or less from horizontal.  
3) Fuse values above 20A have additional spacing requirements; see the user's manual for details.  
Yaskawa Solectria Solar does not supply optional fuse sizes.



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Document No. FL.PVI25TL-208.01 | 07/17/2024 | © 2021 Yaskawa America, Inc.

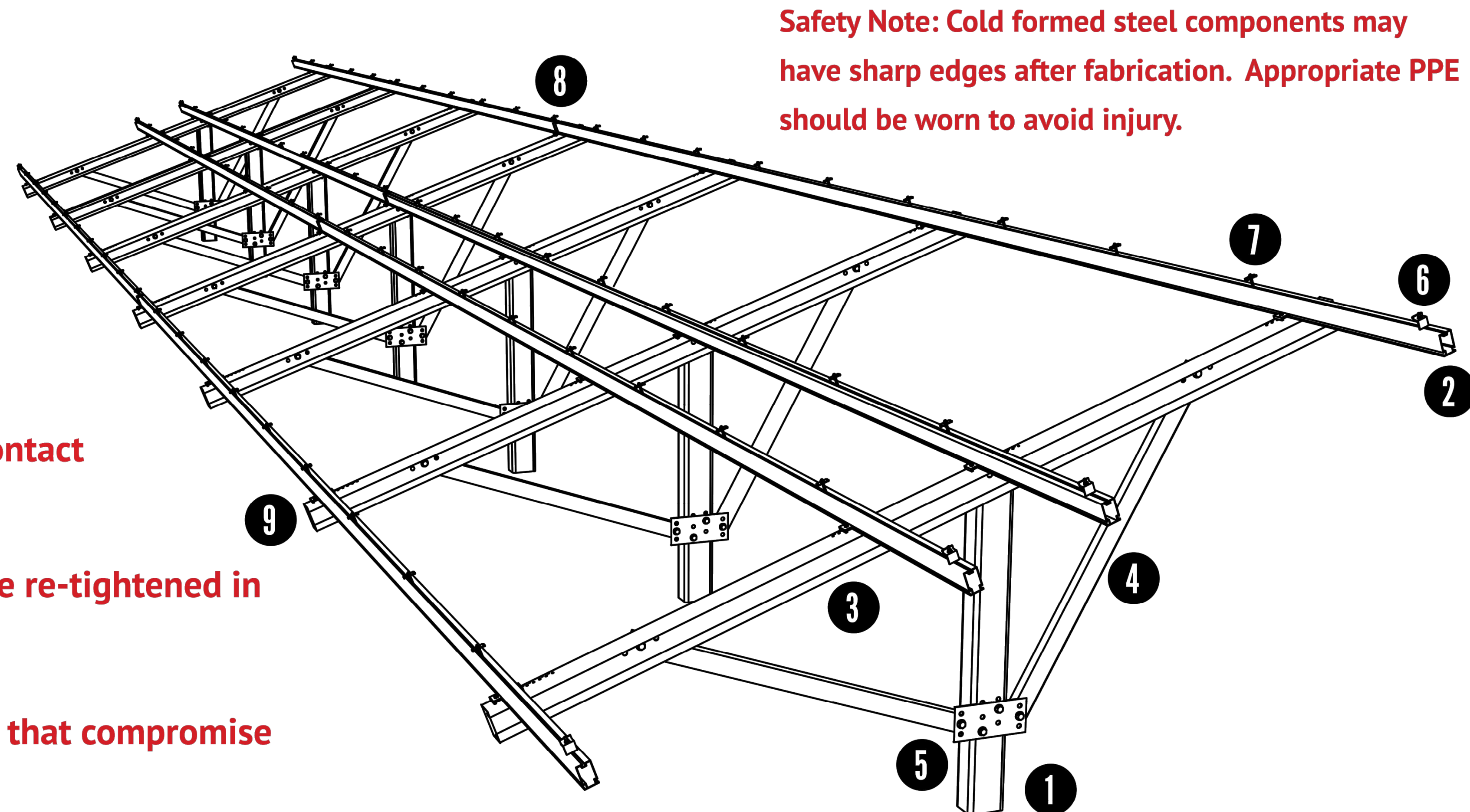
#### Safety Notes:

Cold formed steel components may have sharp edges after fabrication. Appropriate PPE should be worn to avoid injury.

Load ratings are project specific - please contact Unirac or refer to U-Builder.

Any loose components or fasteners shall be re-tightened in accordance with these instructions.

Any components showing signs of damage that compromise safety shall be replaced immediately.



Safety Note: Cold formed steel components may have sharp edges after fabrication. Appropriate PPE should be worn to avoid injury.

| ITEM | COMPONENT                        | MATERIAL                                                   |
|------|----------------------------------|------------------------------------------------------------|
| 1    | Roll- Formed Steel Pile          | 4" or 4.5 " x 6" C Shape (Length Varies by Project)        |
| 2    | Aluminum East-West Beam          | Aluminum Beam with Continuous Slots for Adjustability      |
| 3    | Roll-Formed Steel Top Chord      | C Shape with Custom Hole Pattern for Adjustability         |
| 4    | Roll-Formed Steel Diagonal Brace | C Shape                                                    |
| 5    | Steel Diagonal Brace Plate       | Steel Plate with Custom Hole Pattern for Adjustability     |
| 6    | End Clamp                        | End Clamp Assembly with T-Bolt                             |
| 7    | Mid Clamp                        | Mid Clamp Assembly with T-Bolt                             |
| 8    | Nested Splice Member             | Internal Aluminum Splice Retained with Self-Tapping Screws |
| 9    | East-West Beam Clamp             | Aluminum Extruded Clamp with Stainless Steel Hardware      |



**SKU:** IPC-4/0-6

**UPC:** 0783669532692

**Features & Benefits:**

**Description:**KUP-L-Tap Insulation Piercing Copper Connector, Dual Rated, Main Conductor Range 4/0-4, Tap Range 6-14, Tin Plated, UL, CSA

**Specs**

**Additional Class Attributes**

|                      |            |
|----------------------|------------|
| Connection           | Mechanical |
| Contains DE-OX®      | No         |
| Number of Conductors | 2          |
| Number of Poles      | 1          |
| Number of Ports Tap  | 1          |
| Wire Orientation     | Parallel   |

**Dimensions**

|                        |              |
|------------------------|--------------|
| Drive Size Main        | 1/2          |
| Drive Type Main        | External Hex |
| Height (IN)            | 1.928        |
| Height BX (IN)         | 1.75         |
| Height (mm)            | 48.9712      |
| Length (IN)            | 1.1179       |
| Length (mm)            | 28.39466     |
| Max Insulation OD (IN) | 0.615        |
| Max Insulation OD (mm) | 15.621       |
| Weight (LB)            | 0.252        |
| Weight BX (LB)         | 1.28         |
| Width (IN)             | 1.431        |
| Width BX (IN)          | 4.5          |

**Material and Finish**

|                            |          |
|----------------------------|----------|
| Clamping Hardware Finish   | Tin      |
| Clamping Hardware Material | Steel    |
| Conductor Material         | Aluminum |
| Connector Finish           | Tin      |
| Insulation                 | Nylon    |
| Material                   | Copper   |

**Bolt and Screw Sizes**

|                                        |         |
|----------------------------------------|---------|
| Clamping Bolt Size                     | 5/16-18 |
| Number of Wire Binding Screws per Port | 1       |
| Wire Binding Screw Size                | 5/16    |
| Wire Binding Torque                    | 156     |

**General**

|                             |                 |
|-----------------------------|-----------------|
| Bag Quantity                | 1               |
| Carton Quantity Description | STD PACK QTY 12 |
| Direct Burial               | No              |
| Dual Rated                  | Yes             |
| NEMA Code                   | S-588-4         |
| UPC Code                    | 78366953269     |

**Testing Standards**

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| CSA File Number          | LR-29601                                                          |
| CSA Standard             | C22.2 No. 65-03                                                   |
| Prop65 Compliance        | WARNING: Cancer and Reproductive Harm -<br>www.P65warnings.ca.gov |
| ROHS Indicator           | RES                                                               |
| UL / CULUS Specification | UL 486A/B Listed                                                  |
| UL File Number           | E6207                                                             |

**Environment and Temperature**

|                    |      |
|--------------------|------|
| Temperature Rating | 90°C |
|--------------------|------|

**Color**

|       |       |
|-------|-------|
| Color | Black |
|-------|-------|

**Conductor Ranges**

|                                                        |                 |
|--------------------------------------------------------|-----------------|
| Conductor Range AL Solid (Secondary, tap, load)        | 10 Sol - 14 Sol |
| Conductor Range AL Stranded (Primary, run, main, line) | 4/0 Str - 4 Str |
| Conductor Range AL Stranded (Secondary, tap, load)     | 6 Str - 14 Str  |
| Conductor Range CU Solid (Primary, run, main, line)    | NA              |
| Conductor Range CU Solid (Secondary, tap, load)        | 10 Sol - 14 Sol |
| Conductor Range CU Stranded (Primary, run, main, line) | 4/0 Str - 4 Str |
| Conductor Range CU Stranded (Secondary, tap, load)     | 6 Str - 14 Str  |
| Conductor Range AL Solid (Primary, run, main, line)    | NA              |

**Voltage**

|                |       |
|----------------|-------|
| Voltage Rating | 600 V |
|----------------|-------|

**Additional Technical Specifications**

|                                                         |          |
|---------------------------------------------------------|----------|
| Connector Amperage Rating AL (Derated where applicable) | 60 Amps  |
| Connector Amperage Rating CU (Derated where applicable) | 75 Amps  |
| Conductor Type                                          | Stranded |